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Research Article

Combining in situ synchrotron X-ray imaging and multiphysics simulation to reveal pore formation dynamics in laser welding of copper

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ABSTRACT

Laser beam welding has emerged as a powerful tool for manufacturing copper components in electrical vehicles, electronic devices or energy storage, owing to its rapid processing capabilities. Nonetheless, the material's high thermal conductivity and low absorption of infrared light can introduce process instabilities, resulting in defects such as pores. This study employs a hybrid approach that combines in situ synchrotron X-ray imaging with compressible multiphysics process simulation to elucidate pore-forming mechanisms during laser beam welding of copper. High-speed synchrotron X-ray imaging with an acquisition rate of 20,000 images/second facilitates the identification of relevant process regimes concerning pore formation during laser beam welding of copper with a wavelength of 1070 nm. Furthermore, in situ observations with high temporal and spatial resolution present a unique database for extensive validation of a multi-physics process simulation based on welding processes using different concentric intensity distributions. These validated simulation results enable thorough comprehension of process-related pore formation based on the interaction between keyhole, melt pool and resulting flow field. The findings show that pore formation is driven by four different mechanisms: bulging, spiking, upwelling waves at the keyhole rear wall and melt pool ejections. The synergy of high-speed synchrotron X-ray imaging and multi-physics modeling provides a fundamental understanding of the chronological sequence of events leading to process-related pore formation during laser beam welding of copper.

1. Introduction

The laser beam has become a key technology, particularly in the fields of additive manufacturing (AM) and welding, due to its high intensity, good focusability and non-contact energy input. Its directed and locally limited energy input allows the processing of metallic materials in different regimes ranging from conduction mode to deep penetration keyhole welding. Conduction mode welding is characterized by heat conduction, where the laser beam is absorbed at the material surface, resulting in a melt pool with an almost equal depth and width. To achieve keyhole mode, a higher laser intensity has to be used, causing the material to vaporize in addition to melting. This results in the formation of a characteristic keyhole, a vapor channel surrounded by molten material. The laser beam undergoes multiple reflections and absorptions inside the keyhole, leading to higher energy deposition in the workpiece compared to conduction mode. However, this also results in transient absorption conditions. In particular, the local evaporation-induced recoil pressure leads to a continuous change in the keyhole surface, affecting the angle of incidence of the laser beam (a decisive factor influencing the degree of absorption [1]) as well as the resulting proportion and angle of the reflected beam component. The local and temporal fluctuations of the energy input are accompanied by a highly dynamic momentum and energy transfer between the evaporation zone and the melt pool. This leads to non-harmonic oscillations of the keyhole with azimuthal, radial and axial modes as shown in

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[2] and [3], affecting the local pressure equilibrium as well as the temperature field [4]. This results in transient process conditions that can have a significant impact on the resulting quality and component properties, e.g., temperature gradient and solidification rate affect the resulting metallurgy [5], fluctuations in the keyhole lead to irregular weld depths and spiking (abrupt increase in welding depth) [6], or result in imperfections such as spatter and pores [1,7]. Pore formation is of particular significance since porosity represents a hidden defect, meaning it is not visible from the outside. Considering AM and welding technology, pore formation has been extensively studied in recent years, leading to the description of various pore formation mechanisms. Across materials and processes, the formation of pores or trapped voids can be divided into lack of fusion flaws, gas porosity, and pores due to melt pool instabilities. Among these, the latter can be caused by keyhole dynamics [8].

Considering previous experimental studies, high temporal and spatial resolution in-situ X-ray analyses were performed to elucidate pore formation processes within the interaction zone between the keyhole and the melt pool. Observations revealed the pinching off of pores in the area of a needle-like keyhole bottom [9], the expansion of the keyhole by vaporization due to reflected laser radiation in the lower rear region and its fractional pinch-off [10], or a partial collapse of the keyhole in the area of the rear wall [10]. The resulting pores contain metal vapor and ambient gas in particular and are frozen upon contact with the solidification front [10]. Their volume usually decreases as a result of volume contraction during cooling and condensation of metal vapor [11]. The time scales described for pore formation range from microseconds in processes such as laser powder bed fusion, where keyholes have relatively small depths and high processing speeds (e.g., keyhole depth of about 400 μm at a processing speed of 0.45 m/s [12] to milliseconds in welding processes where keyhole depths exceed 3 mm at a processing speed of 1 m/min ($0.01\overline{6}$ m/s) [13].

To gain deeper insights into the mechanisms underlying pore formation across various scenarios, experimental studies were enhanced through physics-based simulations, which offer further insights into physical systems under idealized conditions. They allow direct observation of parameters that may be challenging to achieve in experiments at the desired length- and time scales, providing valuable support for experimental observations in the investigation of cause–effect relationships [14].

An early numerical investigation into process-related pores in deep penetration laser welding was conducted by Pang et al.[15]. They utilized the level set method to track the free surface of the keyhole during welding of a titanium alloy. Their model accounted for the effect of the gaseous phase by imposing boundary conditions on the free surface, where the recoil pressure is a function of temperature only. Their observation of a low surface temperature inside pinched-off bubbles, which subsequently persisted as pores, led to the hypothesis that a rapid decrease in pressure must occur within the bubble. This decrease in pressure could facilitate the entry of shielding gas into the pore, where it remains trapped. However, verification of this hypothesis was not conducted due to the exclusion of gas phase dynamics, evaporation and condensation from the model. Lu et al.[16] employed similar model assumptions to simulate pore formation in deep penetration welding of steel. They linked the formation of pores with the creation of a liquid bridge that seals off the lower section of the keyhole. They attributed this phenomenon to higher flow velocities at the keyhole wall, particularly at higher laser powers, thus promoting porosity. Additionally, they addressed the role of the solidification rate, noting that a larger melt volume enhances the chances of a bubble escaping the melt pool instead of remaining entrapped as a pore. While the authors analyze the effect of different process parameters on the occurrence of pores, they do not provide insights into the underlying cause–effect relationships other than the increase of large melt flow velocities at the keyhole walls in the case of higher power density. Otto et al.[17] simulated partial penetration welding of both aluminum and copper (separately), employing a more comprehensive model that omitted the common incompressibility assumption and incorporated gaseous phases, distinguishing between ambient gas and metallic vapor. Their findings, in agreement with experimental results, demonstrated the reduction of porosity in copper welding at higher feedrates, at the cost of increased humping. Furthermore, they observed pore formation in aluminum during spiking, where the keyhole partly collapsed after spike formation, sealing off the lowest part of the keyhole which then rapidly solidified resulting in pore formation.

Recently, there has been notable increase in studies of pore formation in laser-based processing, particularly under process conditions encountered in AM through laser powder bed fusion. In this context, a widely employed simulation tool is the commercial software FLOW-3D. Similar to previous studies such as [15] and [16], FLOW-3D does not include gaseous phases but accounts for their effect on the liquid via imposed boundary conditions. Evaporation-induced recoil pressure is a function of keyhole surface temperature and can be adjusted to match processing conditions via a coefficient designed to mimic the local pressure conditions and resulting condensation efficiency [18]. Example studies utilizing FLOW-3D include the work of Cheng et al.[19] who studied the formation of pores in stainless steel. They observed large vortex structures in the melt flow, which could pinch off the lowest part of the keyhole and potentially entrap a pore if the recoil pressure fails to re-open the keyhole quickly enough. However, they did not provide further explanation regarding how the melt flow patterns leading to pore formation originate. Furthermore, they attribute the cause for irregularly shaped pores to drag forces when passing the solidification front. Wang et al.[20] compared FLOW-3D simulations with corresponding in-situ X-ray observations of AM for a titanium alloy, attributing porosity to the formation of a bubble due to a local decrease in recoil pressure on the rear keyhole wall following the formation of a small secondary keyhole on the front wall. This led to a rapid forward motion of the melt pool, which locally pinched off the bottom part of the keyhole. Furthermore, they correlated the likelihood of a bubble remaining at the bottom of the melt pool, and thus forming a pore, with the bubble size. Larger bubbles were observed to leave the melt pool more easily due to their greater buoyancy, while the high-speed melt flow below the bubbles pulled them downwards following Bernoulli's principle. Guo et al.[21] utilized FLOW-3D to simulate powder bed fusion of aluminum. The high reflectivity and thermal conductivity of this material allow the observation of pores generated through the keyhole penetrating the solid substrate directly followed by instant freezing. Notably, there is significant discrepancy in the size and shapes of the pores found in the simulation compared to the corresponding X-ray data used for validation. This disparity could potentially stem from neglecting gas phases and simplifying evaporation phenomena in the modeling approach. Although copper has an even higher reflectivity and thermal conductivity compared to aluminum, and the considered keyholes in the present study are more than twice as deep, the mechanisms observed in [21] share qualitative similarities.

Higher-fidelity modeling approaches have also been employed recently to investigate pore formation in AM. One example is the work of Leung et al.[11], who also include gas phases in their model, distinguishing between ambient gas and metallic vapor. However, they utilized a phenomenological recoil pressure model akin to those discussed previously. This model employs a constant condensation efficiency coefficient and explicitly incorporates the recoil pressure's contribution to the momentum equations on both the gas and liquid sides of the interface. By distinguishing between ambient gas and metal vapor, they show that the bubbles mainly filled with metal vapor would shrink significantly and, if they persisted, they would contain a vacuum. On the other hand, bubbles filled with argon would not shrink as significantly and would remain filled with argon. Furthermore, they argue that pore formation is less likely to occur if the keyhole bottom is filled with argon only, and instabilities primarily arise when some metal vapor is also present at the keyhole bottom. This contrasts with previous hypotheses suggesting that ambient gas being pulled in the keyhole bottom is a key component of pore formation. They also argue that a high concentration of metal vapor inside the keyhole will shield the keyhole from incident laser energy and hence promote an inhomogeneous distribution of laser energy, serving as the initial perturbation that eventually leads to the pinch-off of the keyhole bottom. Yu and Zhao [22] also utilize a more comprehensive modeling approach to simulate AM of a titanium alloy. Their model includes the gas phase, but they do not distinguish between different gaseous species. While they also explicitly calculate a phenomenological recoil pressure and add it as a source term to the momentum equations, they employ a dedicated model for both evaporation and condensation, respectively, in contrast to accounting for condensation via a constant evaporation efficiency. They provided a detailed explanation of the onset of a J-shaped keyhole, followed by necking, pinch-off and the subsequent formation of a pore. Furthermore, the authors quantitatively compared surface tension force and recoil pressure, demonstrating that their imbalance is a driving factor in pore formation, while Marangoni currents play a negligible role, at least locally at the keyhole wall. Additionally, they analyzed the role of the condensation rate on the formation of microjets that ultimately split the keyhole to form a pore and drill a new, small keyhole. It is concluded that the condensation rate plays a crucial role in pore formation, highlighting the importance of accurately modeling evaporation and condensation, and the role of gaseous phases.

All the aforementioned numerical studies of pore formation rely on a phenomenological recoil pressure model, where the local recoil pressure induced on condensed matter by the evaporating material depends only on temperature, and an assumption has to be made for the condensation rate. As noted by Flint et al. [23], higher fidelity modeling approaches that omit phenomenological recoil pressure models and calibrated condensation rates are essential if evaporation-dominated phenomena and thermo-capillary stability are of interest, which are precisely the relevant phenomena during pore formation.

The formation of pores during laser welding of copper poses a particular challenge, since the increasing use of copper in electromobility [24], power electronics [25] and battery systems [26], particularly for high electrical conductivity contacts. Achieving the required quality of such components and assemblies, it is essential to consider mechanical, electrical and thermal properties [27] and [28]. Profound attention is dedicated to understanding pore formation during welding of copper materials due to the fact that pores are entrapped within the weld seam [29], consequently decreasing achievable mechanical properties and reducing the effective current-carrying cross-section [30] resulting in thermal losses [24]. Moreover, manufacturing such electrical applications requires large conductor cross-sections. For instance, when welding hairpins for electric motor stators ($\geq$ 3 mm penetration depth, laser beam power up to 6 kW [31]) or busbars ($\geq$ 1-3 mm thickness, laser beam power up to 8 kW [32]). It should be noted that for welding depths $\geq$ 1.5 mm, near-infrared laser beam sources are predominantly used in industry. Laser beam sources with green or blue wavelengths [33], on the other hand, are only used for welding depths $\leq$1.5 mm in industrial applications, as the maximum available power and beam quality are currently still limited.

Although the physical phenomena encountered in laser welding are fundamentally similar to those in AM of steel or titanium alloys, some major differences must be considered: Firstly, the laser welding process results in a notable increase in the depth-to-width ratio of the keyhole by one order of magnitude or more compared to AM. Secondly, the notably higher thermal conductivity of copper compared to materials like steel or titanium, lead to distinct characteristics, such as narrower and shorter melt pools, along with stronger fluctuations in both gas and melt flow. Regarding experimental investigations, there is limited information available on in-situ X-ray investigations of laser beam welding of copper materials. This is primarily due to copper's high density compared to other materials, as copper has a density of 8.96 g/cm^3, which is approx. three times more dense than aluminum, twice as dense as titanium, and approx. 1.15 times more dense than iron. These density differences result in limitations on the achievable acquisition rates. Additionally, the welding processes for copper require a larger field of view due to the addressed penetration depths, and at the same time an increased sample thickness during the experiment to avoid significant heat accumulation caused by the larger keyhole and melt pool area. Studies by Heider [34] and Miyagi and Zhang [13] offered initial insights into process-related pore formation during copper welding, particularly associated with a bulging of the keyhole bottom. The bulging is attributed to the high power density of laser irradiation, which leads to increased evaporation in this region [13]. When this bulge is constricted by melt flow or a sudden interruption of evaporation – such as shadowing of the laser beam by a melt pool ejection – pore formation occurs.

The use of high-energy, brilliant synchrotron radiation significantly enhances the spatial and temporal resolution of high-speed X-ray imaging, making it feasible to visualize faster phenomena [35]. Studies between 1,000 images/second [36] up to 5,000 images/second by Kaufmann et al. [1] focused on keyhole geometry or void formation. However, the actual recording of dynamic processes such as pore formation is only possible to a very limited extent at such frame rates [36]. Thus, high-speed in situ synchrotron X-ray imaging at a frame rate of 20,000 images/second reveals strong fluctuations and bulging occurring at very short times and at various positions of the keyhole without directly leading to pore formation or other weld defects [2]. In addition, Schricker et al. [2] have further broadened the knowledge gain from Heider [34] by showing that bulging usually occurs in conjunction with melt ejections or pore formation. These findings underscore the substantial potential for enhancing the understanding of pore formation mechanisms through the utilization of high-speed in situ synchrotron X-ray imaging.

Since the formation of process-related pores can be linked to the dynamic behavior of melt pool and keyhole, the temporal modification of laser beam power [37] and beam position [38] have been studied to reduce porosity. Additionally, the spatial manipulation of the intensity distribution based on concentric intensity distributions is gaining major importance in the field of industrial applications. Commercially available systems like COHERENT Adjustable Ring Mode (ARM), IPG Adjustable Mode Beam (AMB) or Trumpf BrightLine utilize such technology. Pore formation can be significantly reduced with such systems realizing intensity distributions consisting of a concentrically aligned core and ring beam as shown by Bocksrocker et al. [39]. However, a fundamental understanding of the mechanisms of action, their interaction with concentric intensity distributions, and a comprehensive description of the phenomena leading to pore formation during laser beam welding of copper is still incomplete.

This study employs a hybrid approach, that integrates in-situ synchrotron X-ray imaging and multi-physics numerical simulation for describing pore formation mechanisms in copper during laser welding. This hybrid methodology allows for the detection and detailed description of various pore formation mechanisms, by considering the interaction between laser beam and copper, as well as the resulting keyhole and melt pool behavior, including their temporal and spatial evolution leading to pore formation. On the experimental side, in-situ X-ray imaging has been carried out using high-energy brilliant synchrotron radiation to achieve a high spatial and temporal resolution for the keyhole and melt pool dynamics in copper (20,000 images/second). The use of sophisticated concentric intensity distributions allows a general statement about their impact on process dynamics and related pore formation in copper, as well as a detailed analysis of specific mechanisms leading to pore formation. Nonetheless, there are limitations associated with the experiment, such as the inability to visualize the influence and distribution of metal vapor, the pressure field and flow field. To overcome these limitations, a high-fidelity modeling approach

Table 1
Chemical composition of EN CW004A [41].

Cu	Bi	O	Pb	Others[a]
min. 99.90 wt.%	max. 0.0005 wt.%	max. 0.040 wt.%	max. 0.005 wt.%	max. 0.03 wt.%

[a] The sum of elements besides Cu is defined as the sum of As, Bi, Cd, Co, Cr, Fe, Mn, Ni, O, P, Pb, S, Sb, Se, Si, Sn, Te, and Zn. Ag is excluded.

is undertaken. In this approach, distinct gaseous phases (ambient gas and vaporized metal) are included in the model domain and treated as fully compressible fluids. Evaporation and condensation are modeled as pressure-dependent processes, adding additional coupling and dynamics compared to all models discussed above. No phenomenological recoil pressure model is needed, as the state transition from liquid to gaseous matter is fully captured, and an increase in volume associated with evaporation implicitly yields a local increase in pressure, i.e., the recoil pressure. Here, the term "implicit" refers to the recoil pressure being an implicit result of the model itself, rather than being accounted for via an explicitly calculated phenomenological model. Furthermore, condensation is directly accounted for within the model, without the need of a calibrated condensation efficiency. The predictive capabilities of this comprehensive modeling approach are evidenced by its ability to reproduce with unprecedented accuracy the characteristic knee in the drill rate and onset of fluctuations during stationary keyhole drilling, as recently presented in [40]. Together, these advanced methodologies shed new light on pore formation mechanisms in copper, broadening the fundamental physical understanding and paving the way for optimizing industrial laser welding processes.

2. Materials and methods

2.1. Materials

The experiments were carried out on Copper EN CW004 A (Cu-ETP). The chemical composition is provided in Table 1. The sample size was 50 mm in height, 150 mm in length and 2 mm in width. The welding position is described in detail in Section 2.3. The top surface was prepared by milling to ensure comparable surface conditions during the laser welding experiments (mean roughness: R_z = 1.54 μm, R_a = 0.43 μm). Thermophysical material properties used within the simulations are provided in Appendix A.

2.2. Laser welding setup

The laser beam welding experiments were performed using a COHERENT HighLight FL8000-ARM fiber laser and a Precitec YW52 processing head. The welding process was carried out in air without the use of a shielding gas, in accordance with standard industrial practice for copper welding. The laser beam source was equipped with an adjustable ring mode (ARM) fiber, which enables the superposition of core and ring intensity profile as depicted in Fig. 1a. The intensity distributions of the core beam, ring beam, and their superposition were quantified using the Primes FocusMonitor (D86 beam width measurement method). The power of core and ring beam for each measurement was selected to ensure that the damage threshold of the measuring device was not exceeded, i.e., a core power P_{core} of 300 W and a ring power of P_{ring} = 300 W was used for the individual measurements (see Fig. 1b). The superimposed intensity distribution was quantified for a comparable peak intensity in core and ring beam (P_{core} = 72 W, P_{ring} = 378 W). Table 2 summarizes the relevant information about the laser welding setup. The weldings were performed under air at atmospheric pressure.

Two sets of experiments were carried out to study different welding regimes and their effect on process-related pore formation. In the first experiment, the power of the core, P_{core}, was ramped down linearly from 3.5 kW to 0 kW and, vice versa, the power of the ring, P_{ring}, was ramped up from 0 kW to 3.5 kW, yielding a constant accumulated

Table 2
Laser welding setup.

Parameter	Symbol	Quantity	Unit
Laser beam power	P	3.5	kW
Wavelength	λ	1,070	nm
Focal diameter (core)[a]	–	89	μm
Outer focal diameter (ring)[a]	–	208	μm
Inner focal diameter (ring)[a]	–	109	μm
Focal position	z_0	Copper surface	–
Welding speed	v_W	10	m min^{-1}

[a] See Fig. 1a.

beam power, P_{acc}, throughout the welding process as shown in Fig. 1c. It should be noted that the power threshold of the laser beam source is not expected to have a noticeable impact on the results obtained when conducting the opposed-ramp experiment. Measurements of the linearly ramped laser beam power over time using the same laser beam source showed no significant discrepancies between the nominal power and the actual power, even at power levels below 10 % of the maximum output power [42]. In the second experiment, only the core beam was used with a constant power P_{core} = 3.5 kW (see Fig. 1d). Varying the core and ring power significantly affects the keyhole and melt pool behavior, i.e., different process regimes can be obtained [2]. This variation allows for the investigation of the influence of different process regimes on pore formation and also provides extensive data for validating the numerical simulation.

2.3. High-speed synchrotron X-ray imaging setup

High-speed synchrotron X-ray imaging was performed at the European Synchrotron Radiation Facility (ESRF) at beamline ID19. A mean photon energy of 60 keV with a beam size of 4 × 4 mm^2 was used for copper welding. A Ce-doped $Lu_3Al_5O_{12}$ scintillator with a thickness of 250 μm from Crytur was lens coupled to a Photron SA-Z high speed camera (5x/0.14 NA, frame rate: 20,000 images/second) resulting in a spatial resolution of 4 μm per pixel for the high-speed images. The propagation distance between sample and detector was 4.5 m in order to benefit from propagation-based phase contrast.

Fig. 2a shows a schematic sketch of the underlying experimental setup. The welds were performed as partial penetration bead-on-plate welds on copper (Cu-ETP/EN CW004 A, see Section 2.1). Details of the welding parameters are provided in Section 2.2. The sample width was 2 mm and the focal position was at the sheet top surface. The processing head was stationary while moving the sample at welding speed. The movement of the sample is visualized by v_M in Fig. 2 while v_W represents the resulting welding velocity. This ensured a constant keyhole position during image acquisition.

Fig. 2b shows a section of the recorded area including keyhole, melt pool and the area above the copper surface to gain information on melt pool swellings and ejections. Image processing was carried out to remove beam path-related artefacts within the field of view and to increase the contrast of the interfaces between the keyhole, melt pool, atmosphere and solid base material. Each corrected frame C_n was calculated via Eq. (1) where W_n represents an individual image recorded during the welding process and $\overline{U}$ is an average image including 500 frames recorded without the welding process but with a copper specimen moving at the welding speed. A high flux is required

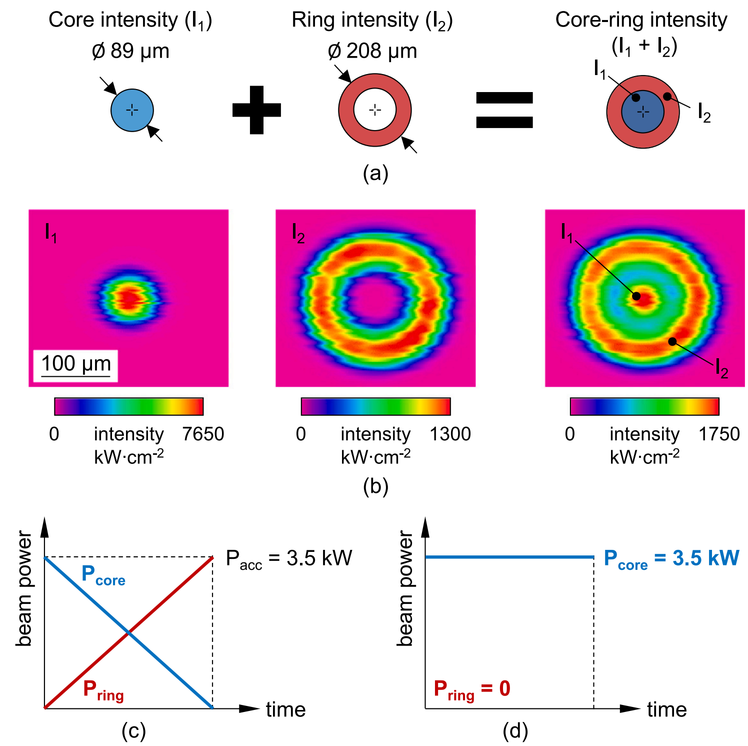

Fig. 1. Description of laser beam parameters: (a) Schematic of superposition of core (I_1) and ring intensity (I_2), (b) individual measurements of different intensity distributions for core intensity (I_1, measured at $P_{core} = 300$ W), ring intensity (I_2, measured at $P_{ring} = 300$ W) and core-ring intensity ($I_1 + I_2$, measured at $P_{core} = 72$ W and $P_{ring} = 378$ W), (c) opposed-ramp of core and ring beam power for first in situ experiment, (d) constant power of core intensity for second in situ experiment, for both experiments the welding feed rate is 10 m/min and the total power is constant at 3.5 kW at 1070 nm wavelength. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

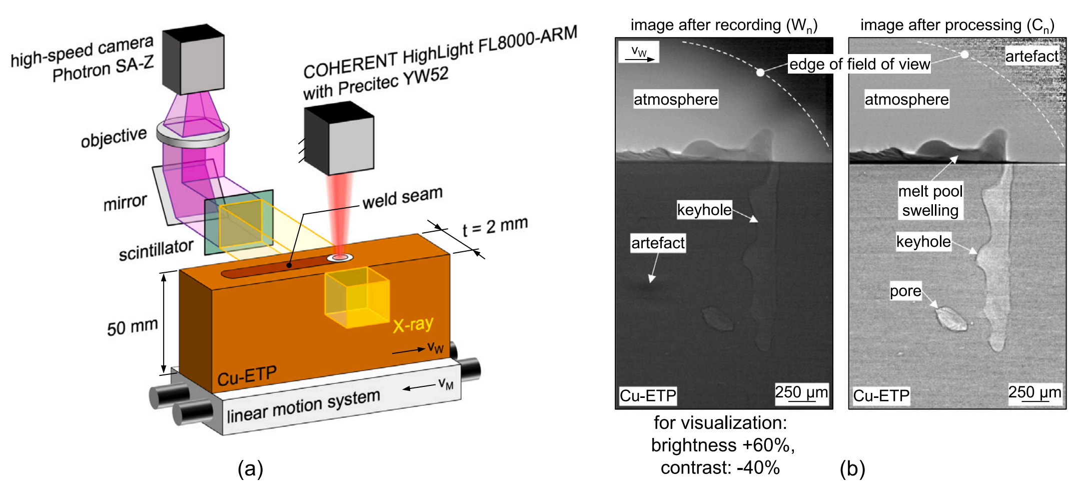

Fig. 2. (a) **Experimental setup** of welding probe and X-ray imaging, b) recorded images before and after image processing.

for high-speed X-ray imaging of copper, resulting in a limited field of view and artefacts in this region after image processing.

$$C_n = \frac{W_n}{\overline{U}} \tag{1}$$

2.4. Simulation model

The coupled multi-physical process is described via a three-dimensional, transient continuum mechanics model identical to that used in [40]. Additional information on the model and its underlying assumptions and equations is provided, e.g., in [43], where humping and the transition to cutting in welding were simulated. A rigorous description of the model's mathematical framework, including a detailed outline of the numerical implementation of the solution algorithm can be found in [40]. Therefore, only a brief description is given hereafter.

Fluid flow of the multi-phase mixture of N dispersed or segregated phases i in homogeneous equilibrium is modeled using the compressible Navier–Stokes equations in the following form,

$$\frac{\partial(\rho \boldsymbol{u})}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}\boldsymbol{u}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} - \boldsymbol{S}_B + \boldsymbol{S}_S + \boldsymbol{S}_D, \tag{2}$$

where $\boldsymbol{u}$, p and ρ, refer to the velocity, pressure and density of the mixture. The viscous stress tensor, $\boldsymbol{\tau}$, contains viscous shear stresses assuming laminar flow. The source terms $\boldsymbol{S}_S$ and $\boldsymbol{S}_B$ contain forces due to surface tension and gravity, respectively. Furthermore, the source term $\boldsymbol{S}_D$ ensures solid body movement restriction using a porous bed model via the Carman-Kozeny equation following the approach of [44].

Notably, no source term for forces due to recoil pressure is included in Eq. (2), as state transitions between condensed and gaseous matter are fully captured within the Mass-of-Fluid model and therefore, the forces resulting from the expansion of newly created vapor are a result of the model itself, and do not need to be accounted for explicitly via a simplified, phenomenological model. The continuity equation is formulated individually for each phase i (where a phase refers to one aggregation state of one material), using the Mass-of-Fluid method [40], as

$$\frac{\partial \rho_i}{\partial t} + \nabla \cdot (\boldsymbol{u}\rho_i) = \dot{\rho}_{i,m} + \dot{\rho}_{i,s} + \dot{\rho}_{i,e} + \dot{\rho}_{i,c}, \tag{3}$$

where the quantity $\rho_i = m_{i,CV}/V_{CV}$ is defined as the local mass density, where $m_{i,CV}$ is the mass of phase i within a control volume and V_{CV} refers to the volume of the aforementioned control volume. The four source terms on the RHS of Eq. (3) contain changes in ρ_i due to melting, solidification, evaporation and condensation, respectively, i.e., the change in a phase's mass due to state transitions (e.g., if liquid phase evaporates, it will lose mass through the respective source term, exactly matching the gain in the respective vapor phase through the respective source term, cf. [40] for an in-depth explanation). The mixture density can then be easily obtained as $\rho = \sum_i \rho_i$. The constitutive relationship between a change in pressure p and the resulting change in a phase's density with respect to its uncompressed (i.e., at reference pressure p_{ref} = 1bar) reference density $\rho_{ref,i} = \rho_i(p_{ref})$ is modeled through the Tait equation [45] as

$$p - p_{ref} = \frac{K_i}{\gamma_i}\left(\left(\frac{\rho_i}{\rho_{ref,i}}\right)^{\gamma_i} - 1\right), \tag{4}$$

where K_i denotes the bulk modulus of phase i and the choice of γ_i yields different behavior, such as that of an ideal gas, or that of a compressible liquid. Analogously to mass conservation, an equation for the conservation of energy is introduced as

$$\frac{\partial H_i}{\partial t} + \nabla \cdot (\boldsymbol{u}H_i) + S_{p,i} = Q_{abs,i} + \dot{H}_{i,m} + \dot{H}_{i,s} + \dot{H}_{i,e} + \dot{H}_{i,c}, \tag{5}$$

where H_i is the energy of phase i and the source term $Q_{abs,i}$ refers to the amount of laser energy absorbed by phase i, where the total amount of energy absorbed at any given point, Q_{abs}, is distributed among phases via $Q_{abs,i} = \rho_i Q_{abs}/\rho$. Note that the value of Q_{abs} depends on the local absorptivity of phases, hence, if vapor absorptivity is set to zero (as done within the present study, cf. Table 8), Q_{abs} will accordingly be zero in regions where only vapor is present. The source $S_{p,i}$ contains terms related to thermal-pressure coupling and the last four terms on the RHS of Eq. (5) represent changes in H_i due to melting, solidification, evaporation and condensation, respectively, analogously to the RHS sources in Eq. (3). Heat conduction is modeled (with c_p and λ denoting heat capacity and thermal conductivity of the mixture) through an additional equation as

$$\frac{\partial (\rho c_p T)}{\partial t} = \nabla \cdot (\lambda \nabla T). \tag{6}$$

Heat conduction cannot be directly included in Eq. (5), because of it being a temperature-driven process, but the temperature field related to a certain distribution of H_i is generally not known for $i > 1$, and therefore a decoupled strategy in the manner of temperature recovery methods [46] is employed. Evaporation is modeled through the Hertz-Knudsen model as

$$\dot{\alpha}_v = \frac{1}{\xi \rho_{ref,l}}\sqrt{\frac{M}{2\pi RT}}\left(p_{sat} - p\right), \tag{7}$$

where α_v is the volume fraction of metal vapor, and M, R, $\rho_{ref,l}$ and ξ denote the molar mass, universal gas constant, temperature-dependent reference density of liquid metal and the liquid-vapor interface thickness, respectively, with ξ taken as the Finite Volume cell size normal to the liquid-vapor interface. The saturation pressure p_{sat} is obtained from the Clausius-Clapeyron equation, including a Watson correction [47] in the material's latent heat of vaporization to account for the decrease of latent heat towards the material's critical point. Condensation is modeled analogously to Eq. (7). As outlined above, recoil pressure is an implicit result of the phase change following evaporation, as mass transfer from liquid to vapor state is associated with an increase in volume and thus pressure, and therefore no explicit model is needed. Energy-based melting and solidification is modeled analogously as a transfer of mass and energy between the respective phase-pair. The amount of absorbed laser energy, Q_{abs} is obtained through first solving a set of two coupled radiative transfer equations (RTE) for the fraction of laser intensity crossing gaseous and condensed matter, respectively (cf. [40] for more details on the mathematical structure of these RTEs). Through this method, the exact intensity distribution and three dimensional caustic of the laser is considered in the beam propagation. After the first incidence of the laser beam on condensed matter, subsequent reflections (and absorption) are calculated through a ray tracing algorithm, where the local angle between the incident laser beam and the interface between gaseous and condensed matter is used to calculate reflections. For a more detailed description of the beam propagation model, the reader is referred to [40,43]. The fraction of reflected energy, r, is calculated under consideration of the local angle of incidence, θ, via the Fresnel equations,

$$r_S = \left|\frac{cos(\theta) - \underline{n}\sqrt{1 - sin^2(\theta)/\underline{n}^2}}{cos(\theta) + \underline{n}\sqrt{1 - sin^2(\theta)/\underline{n}^2}}\right|, \tag{8}$$

$$r_P = \left|\frac{\underline{n}cos(\theta) - \sqrt{1 - sin^2(\theta)/\underline{n}^2}}{\underline{n}cos(\theta) + \sqrt{1 - sin^2(\theta)/\underline{n}^2}}\right|, \tag{9}$$

$$r = \frac{r_S + r_P}{2}, \tag{10}$$

with $\underline{n} = n + \mathrm{i}\kappa$, where n and κ denote the material's refractive index and extinction coefficient, respectively, and i is the imaginary unit. The mathematical framework laid out above is discretized using the Finite Volume Method, using the C++ continuum mechanics library `OpenFOAM` [48]. For information on the Finite Volume Method and its implementation in `OpenFOAM`, the reader is referred to [49]. A brief overview of the solver algorithm employed within each time step of the transient solution procedure is laid out in Table 3.

For an in-depth validation of the above-presented modeling approach and its numerical implementation, the reader is referred to [40], where also keyhole welding of copper is simulated and compared to in situ X-ray observations of Schricker et al. [2], showing excellent agreement even for highly dynamic events like the occurrence of bulging.

Table 3
Overview of numerical solution algorithm within one time step of the transient solution algorithm.

1	**Mass and energy advection**
	Solve Eqs. (3) and (5)
2	**Beam propagation**
(a)	Solve steady-state RTEs for laser intensity to obtain initial (up to first incidence) beam propagation, reflection and absorption
(b)	Employ ray tracing algorithm for subsequent reflection and absorption following initial incidence
3	**Heat conduction**
(a)	Calculate T numerically from given distribution of H_i
(b)	Solve Eq. (6)
(c)	Update H_i from new temperature distribution
4	**Phase change**
	Update ρ_i and H_i following melting, solidification, evaporation and condensation
5	**PISO-Loop**
	Solve pressure-momentum coupling in compressible PISO-loop (following the approach of [50]) to obtain $\boldsymbol{u}$ and p

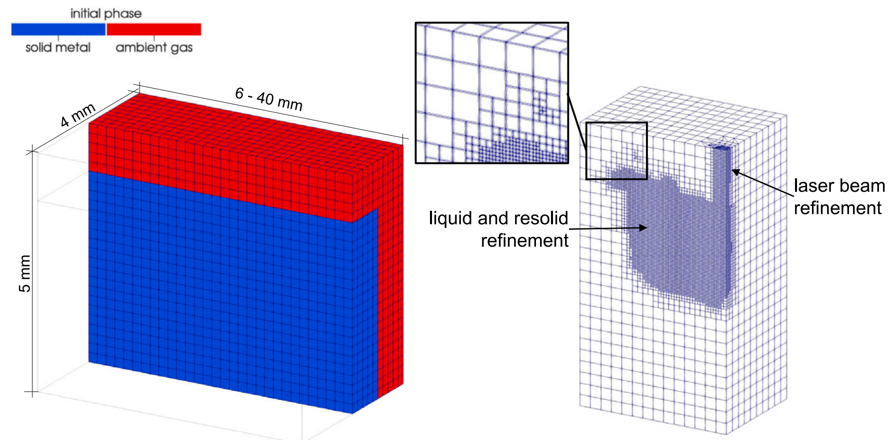

Fig. 3. Computational domain and mesh: initial discretization and phase distribution (left, cut longitudinally), and dynamically refined mesh during exemplary simulation (right, cut longitudinally and transversely). Domain length varies depending on simulated welding length (mesh size according to Table 4). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.5. Simulation setup

In order to reproduce physical phenomena observed in both experiments, in total, four simulations were carried out. Due to the high computational demand of the underlying model, the simulation of the power ramp experiment was split into a start-, mid- and end-simulation, referred to as simulation Sim 1a, Sim 1b and Sim 1c, respectively. The second experiment with constant core intensity (see Section 2.2) was replicated by simulation Sim 2 for an increased length of the weld. The workflow was chosen during the study to reproduce all pore formation mechanisms in the multi-physics simulations (see Section 3.2).

The computational domain, its discretization using hexahedral cells, and the initial phase distribution are summarized in Fig. 3. Dynamic mesh refinement was employed during the simulation, refining the computational mesh depending on the presence of propagating laser intensity, absorbed laser energy and liquid or solidified metal (see Fig. 3, right). The domain boundaries being initialized with ambient gas (air) were set as inlet/outlet boundaries. At the part of the bottom domain boundary initialized with metal, the thermal boundary condition was set to a fixed value of ambient temperature to model the presence of a large heat sink in the experiment (i.e., the remaining copper sheet with its high thermal conductivity, and clamping). At the remaining two boundary patches initialized with metal, the thermal boundary condition was set as zero gradient. Information on the cell size (coarsest/initial and finest cells), simulation dimensions and intensity distributions are listed in Table 4. For simulation 2, a coarser mesh was employed to achieve reasonable computational times given the spatial domain being approximately one order of magnitude larger than the remaining simulations. The time step, Δt, was set following the Courant–Friedrichs–Lewy (CFL) condition [49] of both convection and diffusion, i.e., $\Delta t \leq \min\left(\rho c_p \Delta x^2/\lambda, \Delta x/|\boldsymbol{u}|\right)$. The material properties used for the simulation are provided in Appendix A. The core intensity profile is modeled as a superposition of 50% Gaussian and 50% top-hat distribution with 89 μm diameter (using the $1/e^2$ definition) according to the assumptions of [51] on high power fiber laser beam profiles. The ring intensity profile is assumed to be a top-hat beam. For the sake of repeatability, the intensity profile of the balanced power distribution (50% core and 50% ring intensity, Sim 1b) is plotted in the appendix, Fig. 17. The simulations were conducted on an Intel® Core™i9-12900K processor. Simulation Sim 1a, Sim 1b and Sim 1c were executed on 4 cores each, with a computation time of approx. 3, 8 and 8 days, respectively and simulation Sim 2 took approx. 14 days on 10 cores.

Table 4
Simulation parameters.

Parameter	Sim 1a	Sim 1b	Sim 1c	Sim 2
Laser beam power core P_{core} (kW)	3.5	1.75	0	3.5
Laser beam power ring P_{ring} (kW)	0	1.75	3.5	0
Coarsest cell size (mm)	0.2	0.2	0.2	0.5
Finest cell size (mm)	0.025	0.025	0.025	0.03125
Welding length (mm)	4	8	8	38

2.6. Limitations and synergies of X-ray radiography and simulation

While radiography is invaluable for in situ visualization of welding processes, the technique encounters specific limitations when applied to copper due to its high density. The requisite thin sample geometry (2 mm in this work), necessary for sufficient X-ray transmission and high frequency acquisition, may not accurately reflect typical industrial scenarios and could potentially influence the welding process through enhanced heat accumulation. In addition, the radiography only captures the melt pool when enough molten material for phase contrast is present. This leaves the proper detection of the solidification outline and the metal vapor unobserved. Furthermore, X-ray radiography provides only a two-dimensional projection of the three-dimensional process.

Conversely, the simulation offers a three-dimensional representation but is inherently constrained by computational resolution. Considerably, a finer mesh resolution leads to increased computational demands, as governed by the CFL condition (cf. 2.4) and a higher number of cells, limiting the resolution of small-scale phenomena such as spiking or micro-porosity. Additionally, simulations are also bound by the accuracy of input material properties like surface tension or refractive indices near the boiling point, which are often determined empirically or through numerical identification. Moreover, while simulations can model complex conditions and multiple phases of matter, they typically assume idealized conditions such as homogeneous material properties and stable laser inputs, which may not fully represent real-world variability. Thereby, fluctuative process behavior solely stems from the underlying coupled nonlinear differential equations.

Despite these individual limitations, the integrated use of X-ray radiography and simulation provides a comprehensive toolset for understanding complex process mechanisms and cause–effect relations. X-ray radiography excels in capturing fine physical details and real-time dynamics, offering novel validation possibilities for simulation models. It particularly stands out in accurately delineating the morphology of the keyhole and other critical features, which are essential for

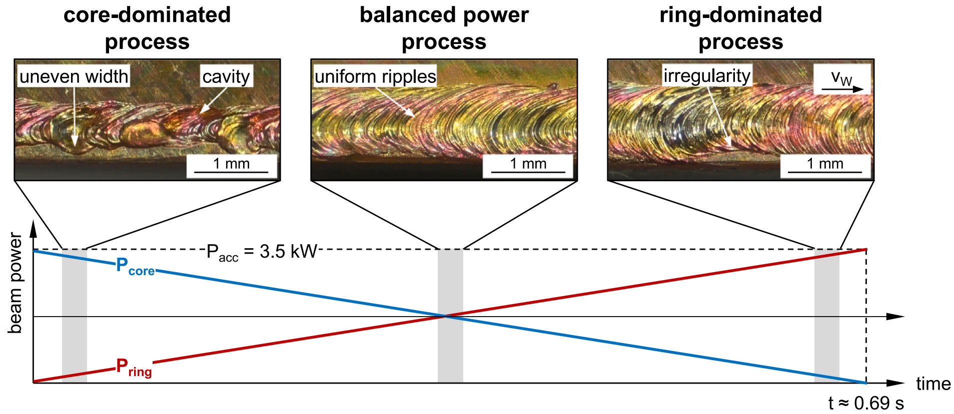

Fig. 4. Sketch of the opposed-ramp experiment with top views on the weld seam. Investigation of weld seam quality for different core ring distributions. Welding feed rate was kept at 10 m/min.

refining simulation parameters and models. On the other hand, simulations complement X-ray observations by providing three-dimensional insights and enabling the visualization of otherwise undetectable information such as temperature distribution, phase changes, laser absorption and fluid dynamic properties like pressure and velocity fields within the weld zone. Significantly, once properly validated with the radiography, simulations can extend to real-world applications without the constraint of sample geometry. This allows for exploring conditions not feasible in experimental setups, making simulations an invaluable tool in industrial scenarios with varying component dimensions and configurations. Furthermore, simulations are notably less expensive than experimental X-ray imaging, particularly for studies involving multiple parameter variations such a power, feed rate, or especially beam shape. The use of validated simulations for optimizing these parameters proves not only cost-effective but also highly efficient, enabling extensive testing and refinement without the need for costly experimental setups.

By merging brilliant radiographic data with robust, validated simulations, this hybrid approach not only overcomes the individual limitations of each method but also significantly enhances our capacity to analyze, interpret and predict the complexities of the laser welding processes.

3. Results and discussion

3.1. Validation of simulation model and description of process dynamics

The changes in process characteristics associated with a variation of core and ring power over time serves as a starting point for validating the simulation. The high speed synchrotron X-ray images provide a large database enabling a validation based on the time- and location dependent keyhole behavior and to describe the process dynamics in the context of pore formation. To comprehend the effect of various intensity distributions, an opposed power ramp was employed. Specifically the core power P_{core} decreases linearly from 3.5 to 0 kW, while simultaneously the ring power P_{ring} increases from 0 to 3.5 kW. Fig. 4 shows the power ramps with corresponding weld surfaces for three different time intervals and intensity distributions. The core power-dominated process results in an uneven weld surface with varying seam width at the top and concavities due to melt ejections. As the core power decreases and the core-to-ring power ratio is balanced, a nearly constant seam width with uniform ripples appears on the surface, indicating a more stable process. In contrast, the ring-dominated process shows slight irregularities on the surface concerning weld width.

Based on these findings, a comparison of keyhole-melt pool interactions is employed to validate the numerical simulations for three different regimes: The core-dominated process, the balanced power process, and the ring-dominated process. This approach allows an extended validation of the numerical simulation, compared to conventional methods which mostly rely on ex situ microsections of the resulting weld bead. The validation approach can be legitimized because many process behaviors can lead to the same or at least similar cross-sections and weld seam surfaces — the keyhole morphology is therefore further up the cause-and-effect chain. At the same time, the simulation's ability to describe the process characteristics is tested over a wide range of keyhole and melt pool regimes. In the following, for three representative regimes of the opposed power ramp experiment, X-ray image sequences are compared to corresponding simulation results. These image series represent the temperature distribution within the melt pool, as well as the base material, the solidified outline (white line) and the keyhole as longitudinal section. Moreover, the full three-dimensional contour of the metal vapor inside and outside the keyhole, as well as the liquid metal above the surface, are shown slightly transparent to maximize the information value of each image. The vapor contours consist of semitransparent iso-surfaces at metal vapor volume fraction steps of 0.1 from 0.1 to 1.0, indicating a metal vapor concentration of less than 10% in regions without vapor representation. These regions are therefore mainly filled with ambient gas (air, in this case). It should be noted that all figures of the simulation in this work use the same temperature scale and the same methodological scaling[1] to display the different phases. Also, the same scale is used for simulation and X-ray images.

Fig. 5 shows a comparison between simulation (Sim 1a) and experiment for the core-dominated process. The narrow keyhole with pronounced bulges and a short melt pool with respect to the welding direction, that has no significant elongation towards the weld surface, is characteristic for this regime. The phase contrast of the small melt pool hinders its clear detection in the X-ray images but its shape was confirmed by additional high-speed recordings from the top side without X-ray. A penetration depth of 1.5–2 mm is reached while the process is prone to melt ejections, keyhole collapses and pore formation. Based on various events at time $\tau = 0$ ms, bulges occur at different positions at the keyhole while minor bulges are moving upwards ($\tau = 0.05$ ms). At this point, it requires to mention the heterogeneous distribution of metal vapor. Especially the conspicuous surplus of vapor filling the minor bulge at $\tau = 0$ ms certainly contributes to the ascending movement of the bulge, which is in fact well captured in the X-ray series. The keyhole geometry and related mechanisms are accurately represented when comparing simulation and X-ray imaging experiment also with respect to the time scale.

The balanced power process (Sim 1b) (see Fig. 6a) reveals large bulges occurring at the keyhole bottom reaching more than half of the

[1] For example the visualization of the vapor.

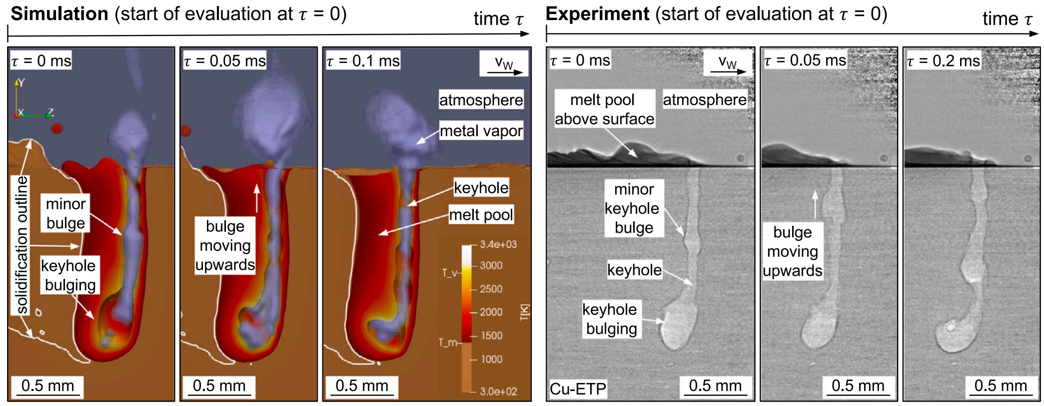

Fig. 5. Validation based on core-dominated welding process: left: snapshots from the core dominated intensity simulation (Sim 1a); right: X-ray images at the beginning of the opposed-ramp experiment. Here, the validation of the idealized simulation is argued upon the excellent resemblance of the keyhole shape in space and time. The characteristics of the core intensity dominated scheme are highlighted by the white boxes. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (~100% core intensity). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

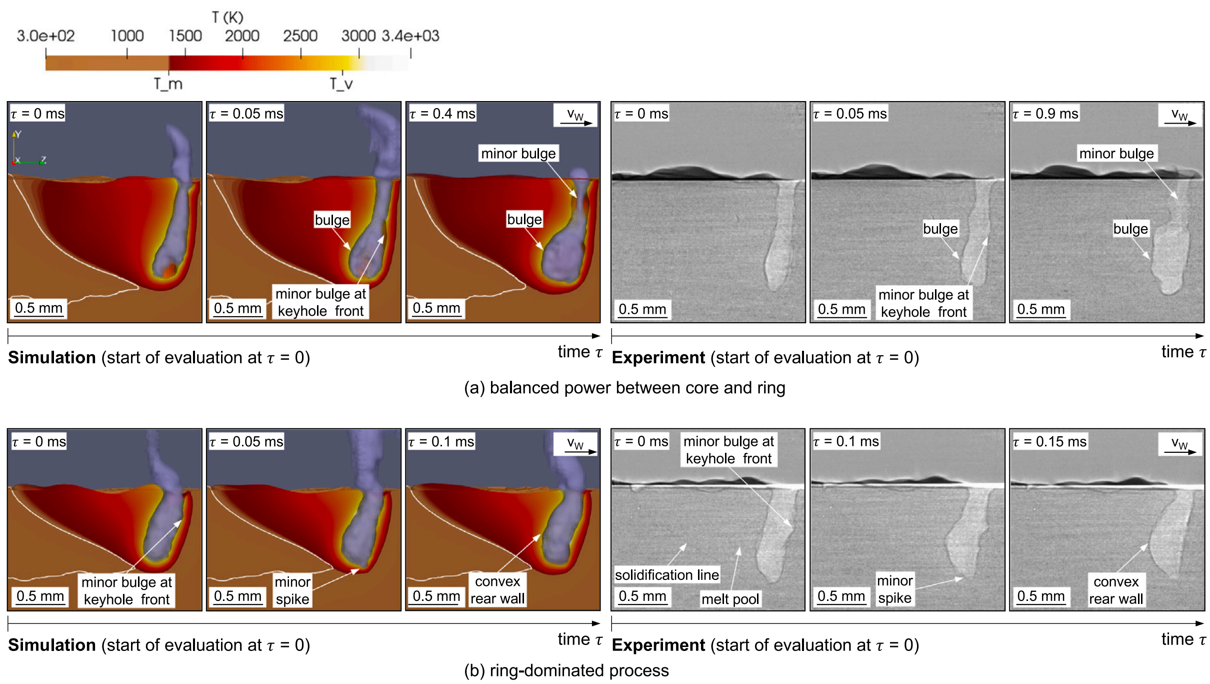

Fig. 6. Validation based on balanced core/ring intensity distribution and ring dominated welding process: Comparison of simulation and X-ray radiography for (a) balanced power between core and ring (simulation (Sim1b)) and middle part of opposed-ramp experiment, and (b) ring-dominated process (simulation (Sim 1c)) and very final part of the opposed-ramp experiment. Here, the validation of the idealized simulation is argued upon the excellent resemblance of the keyhole shape in space and time. The characteristics of both the balanced and the ring dominant scheme are highlighted by the white boxes. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW ((a) ~50% core intensity and ~50% ring intensity; b) ~100% ring intensity. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

keyhole depth and minor bulges can be noticed at the keyhole front while the melt pool is significantly elongated towards the weld surface. The absence of frequent spatter and pore formation suggests a quite constant ratio between opening and closing pressures in order to keep the keyhole stable without pronounced melt ejections by accomplishing a more stable keyhole depth of approx. 1.5 mm. The ring-dominated distribution (Sim 1c) shows a reduced penetration depth of approx. 0.7–1 mm (see Fig. 6b). Even specific characteristics, such as small spikes in the base of the keyhole, are represented by the simulation. This regime in general results in a wider keyhole, often showing a convex rear wall, and a more elongated melt pool in welding direction in relation to the achieved penetration depth. On the one hand, this helps to prevent keyhole collapses in the lower part of the weld. On the other hand, this results in large melt ejections presumably due to the decrease in surface tension pressure working in conjunction with the dynamic and hydrostatic pressures, to close the keyhole. The large melt ejections result in irregularities as shown in Fig. 4. Concerning the metal vapor behavior, both, the balanced process and the ring-dominated process reveal a more homogeneous distribution within the keyhole mitigating the formation of minor bulges.

The simulation represents the laser welding of copper reliably and very well over the different process regimes considered for validation. Specific features and high dynamics (e.g., the movement of minor bulges) are well represented while minor deviations on the time scale between simulation and experiment are negligible due to the correct order of effects leading to certain phenomena. It is therefore assumed that the conditions for the formation of defects and the underlying physical cause–effect relationships can be determined by combining in situ experiments with multi-physics process simulation.

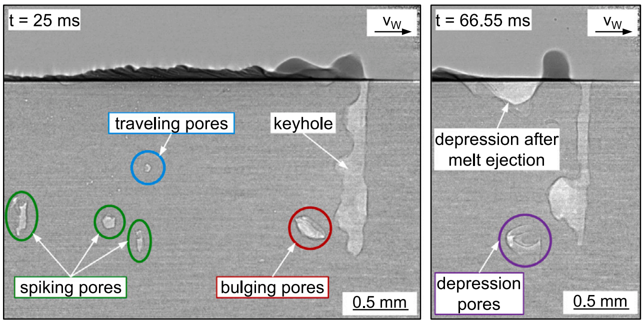

Fig. 7. Exemplary X-ray image showing all four process related pores detected at the core dominated welding regime. Here, both images were recorded at the beginning of the opposed-ram experiment. Process parameters: laser feed rate 10 m/min, wavelength 1070 nm, power input 3.5 kW (~100% core intensity).

3.2. Pore formation mechanisms

The comparison between synchrotron X-ray imaging and respective numerical simulations upon various process regimes has already revealed noteworthy insights into the dynamic behavior of the keyhole and melt pool, which obviously influences the likelihood of process defects. In this regard, the high temporal and spatial resolution of the in situ radiographs enabled the identification of four distinct process-related pore formation mechanisms. Exemplarily, Fig. 7 depicts two X-ray images recorded during the core-dominated regime at 25 ms and 66.55 ms after starting the opposed-ramp experiment. In the following investigation of the underlying mechanisms, differentiation is made between:

- *Bulging pores*: originating from keyhole bulges generated and separated by melt pool convection.
- *Spiking pores*: occurring at the bottom of the keyhole that freeze immediately after a spiking event.
- *Traveling pores*: traveling within the melt pool after being detached from strongly fluctuating keyhole regions.
- *Depression pores*: associated with significant melt ejections as subsequent voids in the weld.

The remarkable agreement between simulation and experiment provides an enhanced capability to study the interaction between keyhole and melt pool and the resulting flow field. Therefore, the combination of X-ray imaging and numerical simulation enables a detailed description of underlying mechanisms for process-related pore formation in laser beam welding of copper.

The corresponding simulation (Sim 1a) allows further understanding of the mechanisms responsible for the formation of bulging, spiking and traveling pores. Since the formation mechanism leading to depression pores was not observed in the short times and weld lengths simulated, a subsequent simulation with significantly increased weld length was performed to increase the chances for various defects to occur that only build up over time and do not occur at high frequencies(Sim 2). In the following, the four pore formation mechanisms will be investigated upon X-ray and simulation snapshots. The representation of the simulation images resemble the one outlined in Figs. 5 and 6 extended by a three dimensional melt pool outline with melt flow plots and the laser beam. These representations should sufficiently explain the underlying mechanisms. However, a more sophisticated representation for each simulation image series can be found in the supplementary material. A detailed description of these additional Figures is attached in the appendix in Fig. 16. Therein, pressure plots and velocity vectors are added. Furthermore, for each X-ray image series, a short video clip is provided in the supplementary material.

3.2.1. Bulging pores

The frame series in Fig. 8 chronologically shows how the first pore in examination originates and emerges. Therein, the first three images in each row reveal the melt flow velocity in vertical and welding directions (denoted by the arrows) in different planes. The white wire frame indicates the liquid metal and the slightly transparent yellow rod shows the laser intensity field until its first interaction with condensed matter. The fourth column pictures the temperature distribution within the melt pool, as well as the base material, the solidified outline (white line), the keyhole and the vaporized metal in and outside the keyhole. The X-ray images on the right column capture the mechanism recorded in the experiment. Considering idealized process conditions in the simulation, temporal disparities in the sequence of events may occur between the simulation and radiography. Therefore, the start of evaluation is set to $\tau = 0$ and the temporal evolution is separately annotated for simulation and experiment. At the onset of pore formation at step 1, the velocity fields generally point upwards indicating a forthcoming melt ejection, and opposed to the welding direction in the lower part supporting local expansion of the keyhole in this region. In step 2, we can see that some droplets form and sporadically detach from the melt pool above the surface of the plate. The remaining and bigger part of the melt is already sloshing back downwards and in welding direction and forms a bulge in the middle, lower section. Subsequently, the small keyhole diameter results in a high closing pressure due to surface tension, which cannot be balanced by the main opening force, which is the evaporation-induced recoil pressure [52] (cf. the partial absence of copper vapor in the simulation result of Fig. 8 at step 2 and the low pressure level revealed in the pressure plot of *S1*). As a consequence, the flow keeps pushing against the bulge which begins to constrict from the keyhole in step 3. Once liquid metal penetrates the bulge, it quickly collapses and detaches, as depicted in step 4. The remaining downward flow causes separation of the elliptical pore from the keyhole which is being formed and remains entrapped in the weld seam as the surrounding material is immediately solidified (step 5). There is only a slight temporal gap of 0.5 ms between steps 3 and 4 in the numerical simulation, whereas, in the experimental setting, the collapsing bulge reconnects to the keyhole through laser beam penetration and eventually collapses at a delayed time instance. Nevertheless, the causal link between the pore formation through bulge collapse and the preceding ascending and subsequent back-flow pattern is evidently established. The investigation of this pore formation mechanism probably reveals the most similarities compared to the one observed by Heider et al. [34] and Miyagi et al. [13] but explains further features leading to the detachment of a pore from a bulge. A conspicuous difference in the present observations is the significantly shorter time intervals of the aforementioned processes. For a more comprehensive insight on the pressure and velocity fields the reader is referred to Figure *S1*. Therein, the pressure levels in the keyhole being lower than atmospheric pressure in conjunction with the gas velocity vectors show how air is sucked into lower parts of the keyhole filling the detached pore. The remaining vapor captured in the detachment instantaneously condensates upon cooling. For a more comprehensive insight on the process dynamics before and after the pore formation event itself, the reader is referred to the video *V1*.

Moreover, the formation of bulging pores is not limited to the lower keyhole regions, namely, the constriction of bulges based on melt pool movement is also feasible for bulges situated in the upper region. As an exemplification, Fig. 9 shows an X-ray image series illustrating a comparable behavior leading to the formation of a pore from an upper bulge. The upward melt flow (step 1) results in melt ejections and supports the growth of a significant bulge in the upper keyhole region (step 2). Consequently, by reversing the flow direction, the constriction process of this bulge is initiated (step 3). As the downward flow persists, the constriction of the bulge proceeds ($\tau = 0.9$ ms) and eventually leads to its detachment (step 4)considerable 0.65 ms after beginning of the constriction. Finally, the pore undergoes solidification upon reaching the solidus front and remains within the weld seam (step 5). For a more comprehensive insight on the process dynamics before and after the pore formation event itself, the reader is referred to the video *V2*.

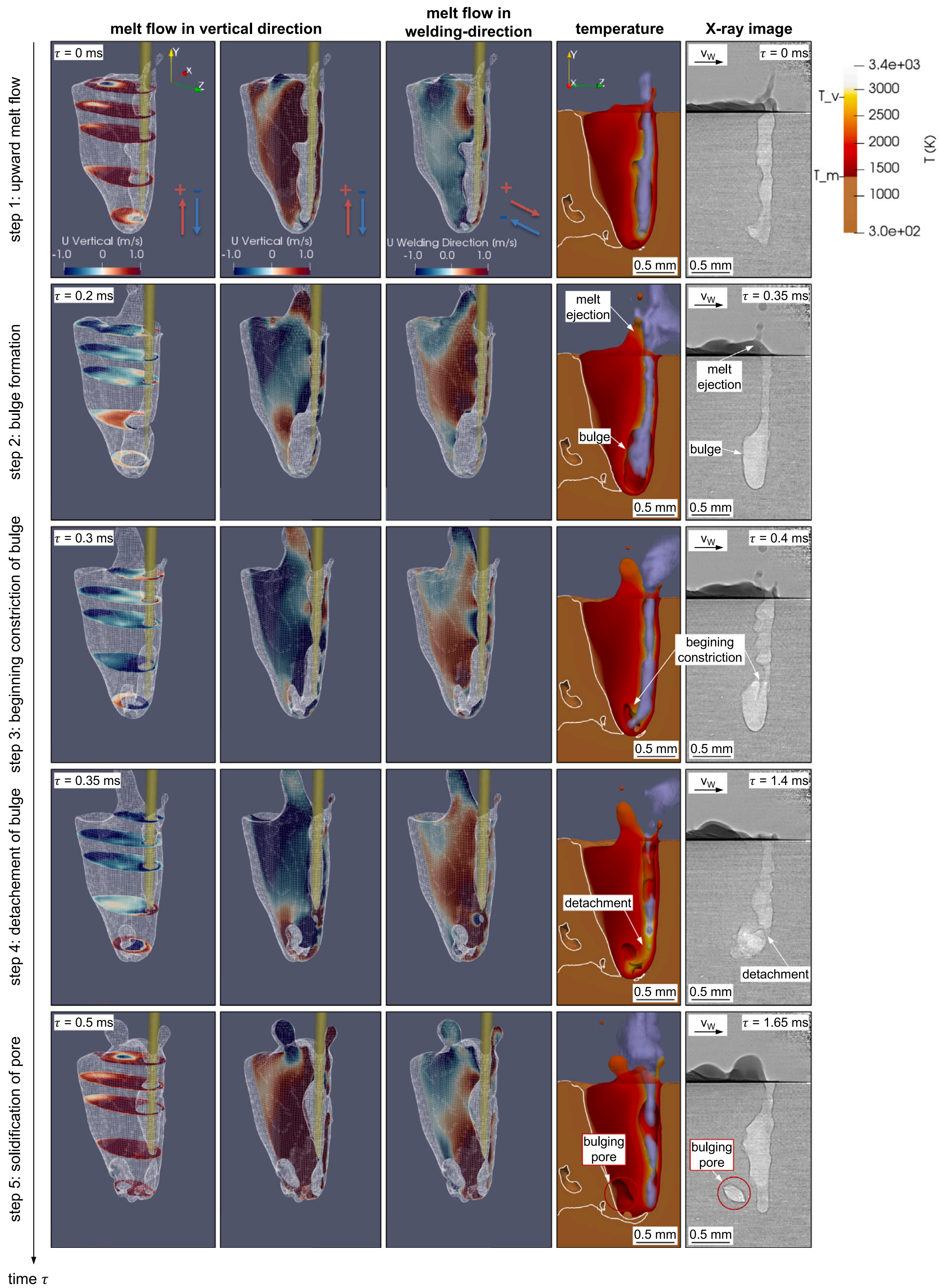

Fig. 8. Bulging pore formation: Evolution of pore formation due to bulging and keyhole collapse (first four columns: core dominated simulation (Sim 1a) revealing the melt pool geometry, the melt flow pattern, the laser intensity until the first incident, the temperature distribution, the metal vapor distribution and the solidification outline; fifth column: X-ray images recorded at the beginning of the opposed-ramp experiment). Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (~100% core intensity). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

3.2.2. *Spiking pores*

The second pore formation mechanism finds its initiation in the spiking phenomenon. Spiking, i.e., the locally limited and rapid increase of the welding depth in the manner of a significant peak, results from a sudden change of the absorbed laser beam intensity in the lower part of the keyhole [6]. This leads to an adjustment of the keyhole geometry [6], which becomes unstable when a critical depth is reached [53]. Similar to Fig. 8, the frame series in Fig. 10 presents the sequence of the events leading to spiking-based pore formation. The two frames on the left side from $\tau = 0$ ms to $\tau = 0.45$ ms depict such a spiking event characterized by a notable increase in penetration. This incident is characterized by the elevated temperature around the tip and a surplus of vaporization indicated by the vapor content inside the spike and also in the increased pressure in the lower part of the keyhole (cf. Figure *S2*). The critical depth of penetration is reached once the vapor pressure exceeds the saturation pressure. Subsequently,

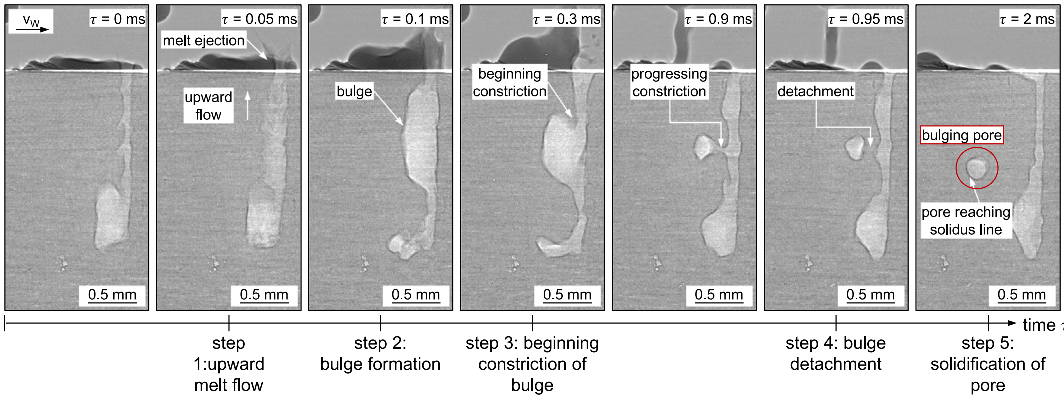

Fig. 9. Bulging pore formation from an upper bulge: X-ray images revealing the evolution of a pore forming from an upper bulge taken from the constant intensity experiment. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (100% core intensity).

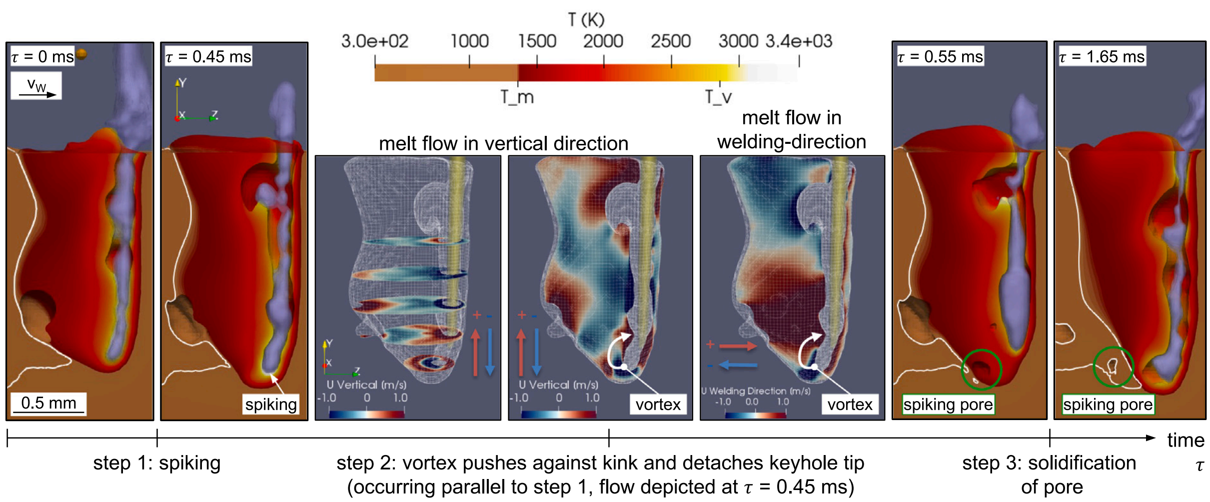

Fig. 10. Spiking pore formation in simulation: Evolution of pore formation due to spiking induced micro vortex. Simulation snapshots taken from the core dominated welding process (simulation Sim 1a) revealing the melt pool geometry, the melt flow pattern, the laser intensity until the first incident, the temperature distribution, the metal vapor distribution and the solidification outline. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (100% core intensity). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

ongoing lasing and possible laser shadowing prevents further heating of the tip which then causes a fast upwards gas stream inside the tip. Additionally, the rapid protrusion causes a micro vortex, visualized by the melt flow fields, which presses against the narrow keyhole, evoking a kink and finally detaching a pore. Due to the rapid increase in penetration depth, the narrow spike tip and its adjacent liquid film approach bulk material of relatively low temperature. Consequently, the pore is swiftly captured by solidification and remains in the weld seam. Notably, the spiking pores observed in aluminum by Otto et al. [17] both in experiment and simulation are not situated at the very bottom of the spike but slightly above it, and they have a more spherical shape than the spiking pores observed in the present study. This difference could be attributed to the lower thermal conductivity of aluminum compared to copper, allowing the spiking pore slightly more time to travel and adopt a spherical shape before freezing.

In the numerical setup, spiking, especially linked with a pore formation, exhibits a relatively limited occurrence, which can be partially attributed to spatial resolution constraints. However, a similar 'minor' spiking pattern can be observed within the X-ray experiment. In particular, the X-ray image series in Fig. 11a shows an abrupt raise in penetration depth until τ = 0.5 ms. The image at τ = 0.95 ms, similar to the simulation, suggests a flow pushing against the kink of the very bottom keyhole, followed by a distinct decrease in penetration depth at τ = 1.05 ms. In contrast to the simulation, the penetration depth remains at a constant level and finally leaves a pore behind.

Because of its inherent origin from keyhole fluctuations, both the frequency and extent of spiking are varying within a certain range. As a demonstration, a second series of X-ray images (see Fig. 11b) shows pore formation during a 'major' spiking event to illustrate the variability intrinsic to this mechanism. Herein, the penetration depth increases by approx. 30% or in absolute values approx. 0.4 mm (Δs) from τ = 0 ms to τ = 0.2 ms in a very narrow shape. The remarkable penetration rate of approx. 2 mm/ms in this time interval suggests rapid drilling into the solid material. This aperture remains connected to the keyhole for another 0.3 ms, as can be seen at τ = 0.7 ms. The rigidity of the drill hole implies the absence of melt inflow refilling the void. In the vicinity of the narrow connection to the keyhole, a certain melt flow and possible condensation appear to exert an upward force dividing the keyhole from the drill hole within the next 0.2 ms, as can be seen at τ = 0.7 ms. Consequently, a pore remains entrapped in the weld seam.

Notably, the diameter of the tip of the spike at τ = 0.2 ms merely measures 50 μm, which is just twice the finest numerical cell length. This particular incident complicates a proper numerical resolution of the spiking event. However, further optimization of the simulation concerning the spatial resolution could potentially support or complement the assumptions captured by the X-ray images in the future. To gain a better overview on the different spiking behavior, the reader is referred to videos *V3* and *V4*.

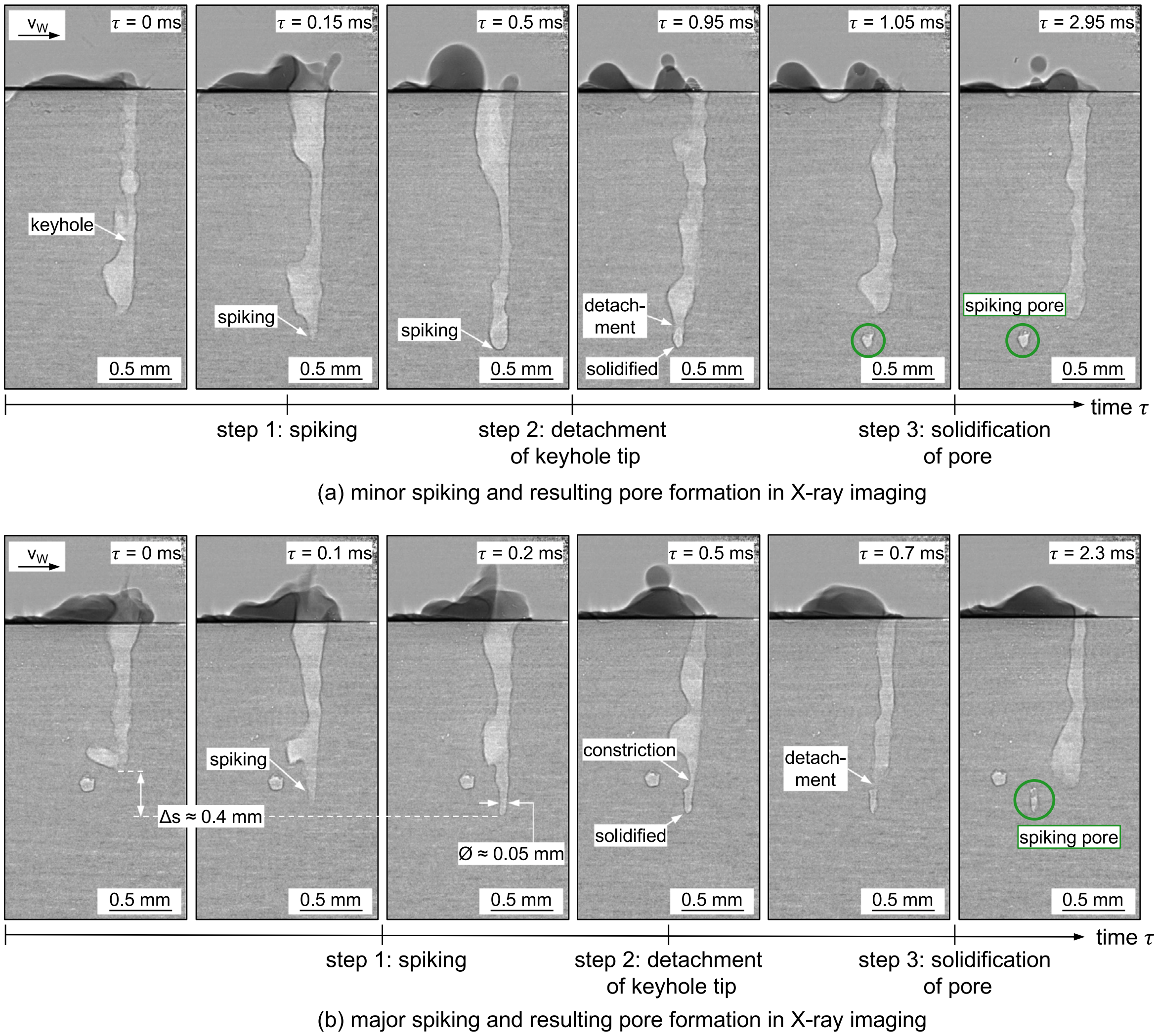

Fig. 11. Spiking pore formation in experiment: X-ray images revealing the evolution of a pore originating from a spiking event taken. (a) **Minor spiking pore:** X-ray images taken constant intensity experiment revealing a moderate spiking event detaching a pore similar to the one captured in the simulation. (b) **Major spiking pore:** X-ray images taken at the beginning of the opposed ramp experiment revealing a much more pronounced and faster spiking event leaving a pore behind. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (~100% core intensity).

3.2.3. *Traveling pores*

The mechanism responsible for the formation of traveling pores incorporates waves traveling upwards at the rear wall of the keyhole. The pronounced dynamics impend a steady bulge trailing in the lower part of the keyhole, giving rise to minor bulges ascending in a wave-like manner. This phenomenon can also be caused by local evaporation and subsequent heterogeneous vapor flows driving liquid metal upwards through interactions at the rear keyhole wall, as observed for welding of ice by Berger et al. [54]. Consequently, the keyhole surface structure exhibits chaotic behavior which in turn manipulates the laser rays and thus, the local evaporation. Subsequently, these dynamics might manifest in a situation depicted in Fig. 12, at $\tau = 0$ ms characterized by a region of high vapor density driving an upwards melt and gas flow (cf. Figure *S3*) at $\tau = 0$ The resulting minor bulge may lead to the detachment of minor pores into the molten material, at $\tau = 0.05$ ms. Subsequently, the pore travels within the melt pool for approx. 0.3 ms when it eventually gets caught by the next minor ascending bulge. The two X-ray images on the right side of Fig. 12 depict the two potential outcomes of a traveling pore: The first scenario involves the escape of entrapment by the keyhole and results in solidification in the rear part of the melt pool, while the second illustrates the reattachment to the keyhole.

Similar to the challenge described in Section 3.2.2, the small length scales of the pore diameters pose a computational constraint for resolving the number of incidents of the mechanism in the simulation. In addition, as the origin of this mechanism is fluctuating in its nature, it also introduces a spectrum of variability in pore size and travel behavior. Therefore, another series of X-ray images (see Fig. 13) provides additional insights into the dynamics of the mechanism. Indeed, this sequence addresses two subsequent detachments of pores. The first pore forms at $\tau = 0.05$ ms from a pronounced ascending bulge and persists within the melt pool for at least 1 ms, where the pore is still surrounded by liquid material, which is faintly perceptible at $\tau = 1.2$ ms. The second pore emerges at $\tau = 1.1$ ms incidentally with the detachment of the entire bulge from the rest of the keyhole. Finally, a few milliseconds later, both pores become trapped within the weld seam by solidification in immediate closeness.

Due to the pronounced dynamic nature of the entire keyhole, the detachment mechanism occurs repeatedly during the experiment throughout the whole penetration depth. To be emphasized is the fluctuating occurrence of the bulge formed at the tip of the keyhole, especially when adopting a pointy shape, making it prone to releasing a pore. Given the narrow melt pool profile in this region, the likelihood of the pore to be caught by immediate solidification is much higher.

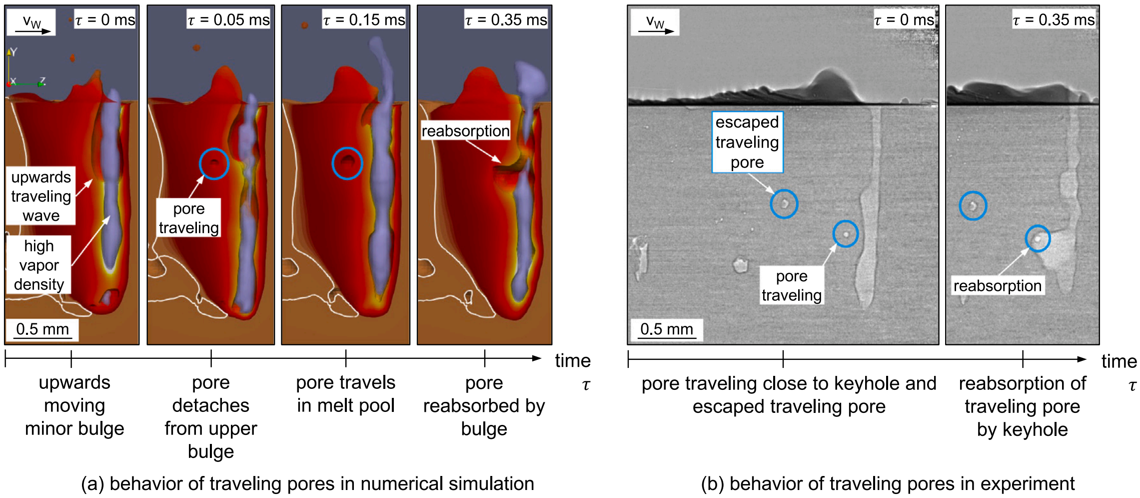

Fig. 12. Traveling pore formation: (a) Behavior of traveling pores in numerical simulation: Simulation snapshots taken from the core dominated welding process (simulation Sim 1a) revealing the melt pool geometry, the melt flow pattern, the laser intensity until the first incident, the temperature distribution, the metal vapor distribution and the solidification outline. In this case, the detached pore is reabsorbed by the next upwards moving bulge. **(b) Behavior of traveling pores in experiment:** Exemplary X-ray snapshots taken at the beginning of the opposed-ramp experiment revealing one escaped traveling pore that solidified and one traveling pore that got reabsorbed by the keyhole bulge. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (~100% core intensity).

To gain a better overview regarding the highly dynamic behavior of this pore formation mechanism, the reader is referred to video *V5*. If compared to bulging pore formation from an upper bulge investigated in Fig. 9 and in *V2*, which also travels some times in the weld pool, the difference in the formation mechanism becomes evident. Whereas the bulging pore finds its initiation in an upwards and subsequent downwards melt flow that constricts and detaches a pore within several ms, the here-described traveling pore stems from the dynamic keyhole morphology change and forms and detaches within the acquisition limit of 0.5 ms.

3.2.4. Depression pores

The frame series in Fig. 14 elucidates the pore formation mechanism observed in the constant intensity experiment and simulation (Sim 2). The conceptual representation resembles the one in Section 3.2.1. Initiated at step 1, the sequence commences with a substantial ejection of molten material, notably more pronounced compared to that depicted in Fig. 8, and entails an expanded upper keyhole structure. As a consequence of the broadened keyhole and the welding speed, the generated vapor has enough space to freely escape the illuminated region which in turn reinforces further evaporation at the absorption front. Within this already intermediate phase of the ejection, the velocity at the upper tip pointing upwards implies a droplet attempting to detach (step 1). In contrast to the experiment, the simulation does not culminate in a detachment of the droplet, yielding the presence of more liquid metal within the forthcoming region of interest. Nonetheless, the fluid flow in the middle and lower part of the melt pool exerts pressure against the kink of the keyhole bottom disconnecting the pore from the rest of the keyhole as evident in step 2. In the following, from step 3 until step 5, a twofold phenomenon arises. On the one hand, the laser beam-induced evaporation and recoil pressure endeavours to restore downward penetration approaching the reattachment of the pore. On the other hand, the depicted liquid flow and the ongoing lasing supports the separation of the keyhole from the pore (step 4). As a consequence, the pore becomes entrapped by solidification merely approx. 0.8 ms after the onset of pore detachment at step 1 and subsequently remains within the weld seam. The less pronounced depression in the simulation can be ascribed to the initially overlooked detachment of the droplet in step 1 along with subsequent material surplus. For a more comprehensive insight into the pressure distributions, the reader is referred to the supplementary Figure *S4*. Therein, in the first 5 snapshots (until τ = 0.85 ms the elevated pressure at the front wall and the slightly subatmospheric pressure at the rear wall yields the sharp absorption front that can be seen in the X-ray images. Since the frame series starts at a very advanced phase of the melt ejection, the reader is kindly referred to the supplementary video, *V6*. This video clarifies the difference to the bulging pore mechanism, outlined in Fig. 8 and video *V1*.

Interestingly, instances of substantial melt ejections, which occur frequently both in experiment and simulation, do not always lead to pore formation. In these cases, either the bottom part of the keyhole is not fully detached by the melt flow, or – if separated – the bottom part is successfully reattached to the new keyhole through drilling. This could also explain why Miyagi et al. [13] reported on significant melt ejections resulting in surface defects such as depressions without finding causal links to pore formation. Indeed, this behavior can be seen at the end of video *V5*.

3.3. General remarks

In addition to the detailed discussions of the individual pore formation mechanisms, some general observations and comments are given below:

- In order to provide a quantitative assessment of pore formation frequency, their occurrences in the constant core intensity experiment upon the track length of 115 mm has been counted. Observations reveal following counts: 6 bulging pores, 9 spiking pores, 32 solidified traveling pores, 17 traveling pores that were reabsorbed by the keyhole, 9 depression pores and 10 depressions without production of a pore. While these counts provide an initial overview, further experiments would be necessary for a robust statistical analysis, and thus, the statistical reliability is limited. Additionally, it has to be mentioned, that the mechanisms of pore formation often overlap, complicating a definitive distinction.
- The present model did not account for direct laser/vapor interactions (by setting the absorptivity and reflectivity of metal vapor to zero). Although we recognize that such interactions must exist, the model effectively captures the process dynamics and pore formation mechanisms without accounting for this effect. Consequently, the assertion by [11], who studied pore formation in a nickel alloy, that pore formation is promoted by higher vapor content in the keyhole could not be confirmed in this study. On the contrary, we suggest that a uniformly vapor-filled keyhole

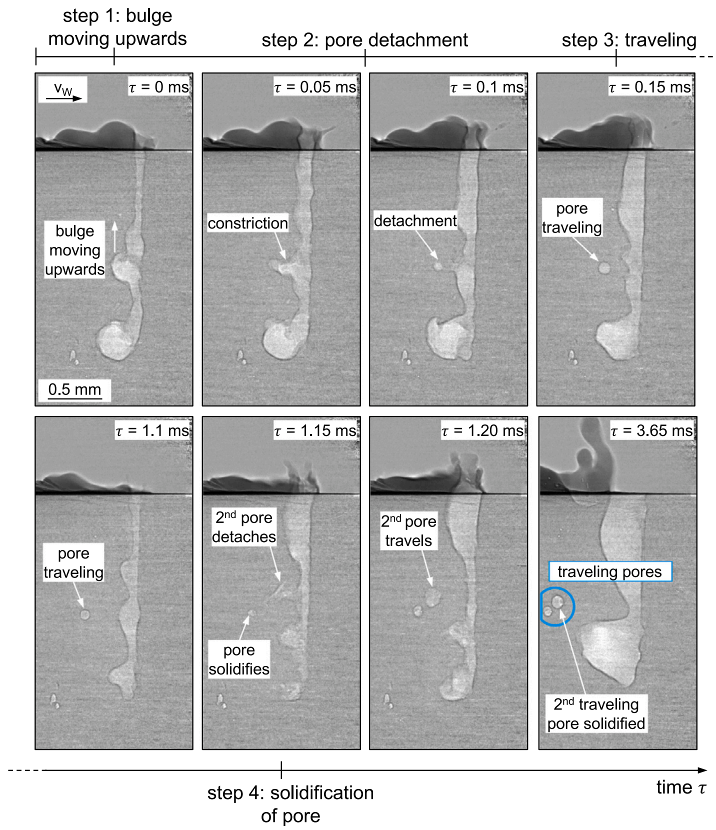

Fig. 13. Traveling pore formation in experiment: X-ray images taken at the constant intensity experiment revealing two subsequent detachments of traveling pores from upwards traveling waves. This sequence shows the extremely short time of the pore detachment and how long the pore may travel within the melt pool. Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (100% core intensity).

may stabilize the process and prevent keyhole fluctuations, as evidenced by the characteristics found in the balanced core/ring intensity regime and the ring intensity regime in Fig. 6. We assume that homogeneous vapor distribution stem from uniform absorption and evaporation throughout the keyhole, resulting in a more stable process.

- The supplementary figures illustrating the distinct pore formation mechanisms consistently reveal keyhole regions with pressures falling below atmospheric levels at some point. Combined with gas velocity plots, these observations indicate atmospheric gas being drawn into the keyhole. Moreover, upon pore detachment, the remaining vapor rapidly condenses, resulting in pores filled with low-pressure ambient gas.
- The morphology of the pores is influenced by both their size and the underlying formation mechanisms involved. Our study identified pointy-shaped spiking pores, in contrast to the spherical pores associated with spiking reported by [17] in aluminum welding (around 1.5 mm). This discrepancy can be attributed to the lower thermal conductivity of aluminum compared to copper, which allows more time for pores to adopt a spherical shape and possibly ascend due to Buoyancy force. More similarities according to the pore morphology are revealed by the pores described by [21] who studied pore formation in aluminum LPBF with about 500 μm penetration depth. However, there is a certain mismatch in the pores observed experimentally and numerically (in the present study and in [21]), which limits direct comparability. Nonetheless, we suggest that high thermal conductivity – as in copper and aluminum – increases susceptibility of pores detaching during a spiking event and that the final pore morphology partially depends on the time available for the pore to take a spherical shape before solidification. This time available is prescribed by the thermal conditions of the surrounding materials and thus, by the application and material itself. Traveling pores typically maintain a spherical shape, either due to their small diameters or their rapid formation. Larger pores, such as depression and bulging pores, exhibit deformation upon solidification, consistent with findings by [19]. Per contra, in this study, these larger pores do not have sufficient time to achieve a spherical shape due to the high thermal conductivity of copper, which leads to rapid solidification.

4. Summary and outlook

This study provides a detailed description and categorization of pore types and formation mechanisms, depending on process conditions, during laser beam welding of copper. A hybrid method combining in situ, high-speed synchrotron X-ray imaging with multi-physics simulations was used to address this challenging task. Both the state-of-the-art 20,000 images/second acquisition rate (for copper) of the radiography and the high level of the numerical model proved to be essential to properly describe the intricate temporal and spatial dynamics of the keyhole and melt pool during laser beam welding. A concentric core-ring intensity profile was used because it allows the determination of relevant process regimes for process-related pore formation in copper and provides data for extensive validation of the multi-physics process simulation. Furthermore, concentric core-ring intensity profiles are currently gaining high interest for industrial applications of copper welding, therefore their detailed analysis is expected to be appreciated not only by researchers but also by welding engineers throughout various industries. A high agreement between simulation and experimental results was achieved considering time-dependent phenomena based on the keyhole and melt pool behavior. The high spatial and temporal resolution allows for the identification of four major process related

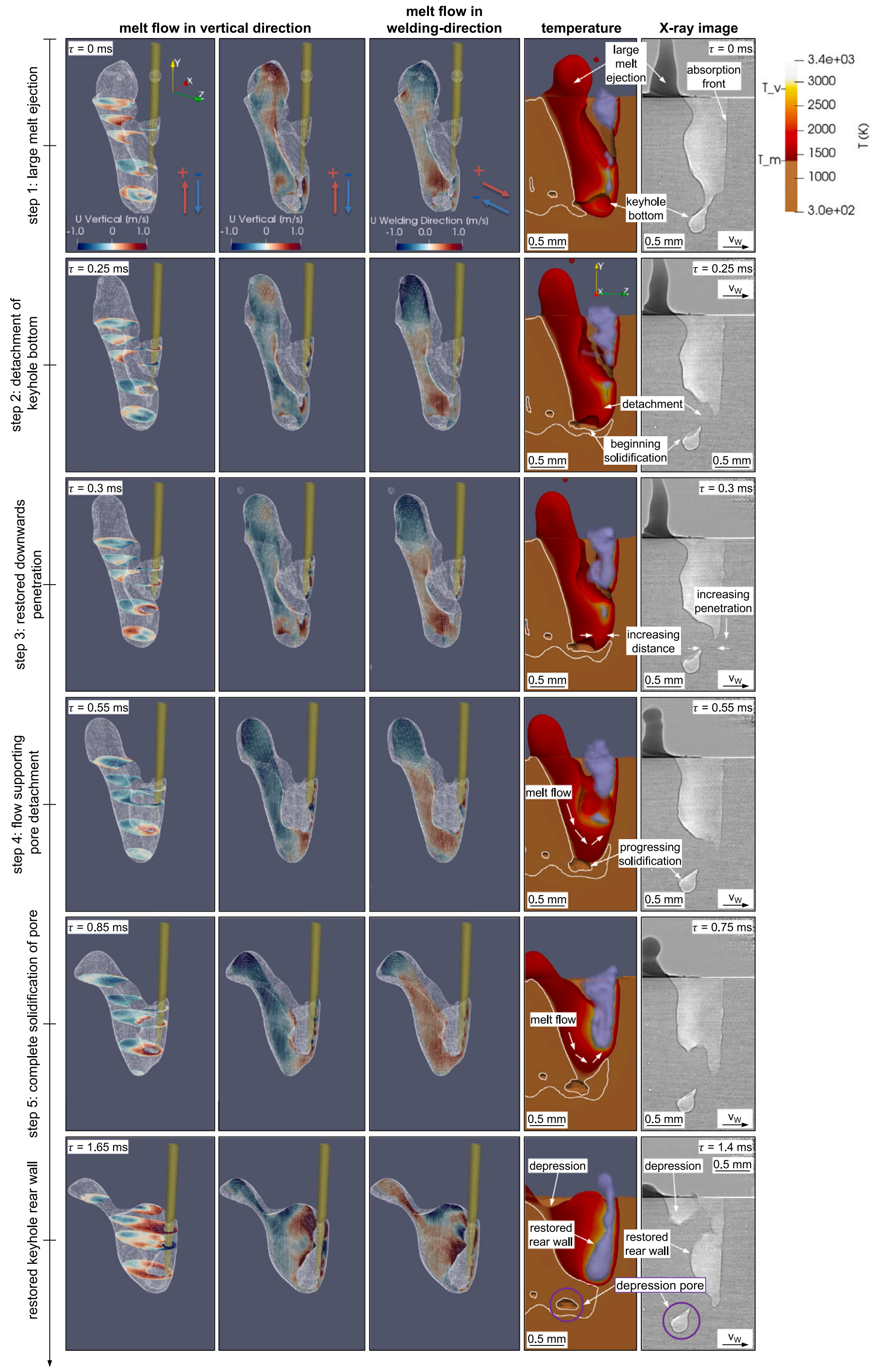

Fig. 14. Depression pore formation: Evolution of pore formation in conjunction with a substantial melt ejection and a remaining depression in the weld seam (first four columns: core intensity simulation (Sim 2) revealing the melt pool geometry, the melt flow pattern, the laser intensity until the first incident, the temperature distribution, the metal vapor distribution and the solidification outline; fifth column: X-ray images recorded at the constant intensity experiment). Process parameters: laser feed rate 10m/min, wavelength 1070 nm, power input 3.5 kW (100% core intensity). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

pore types, schematically illustrated in Fig. 15a-d. In these sketches, the four distinct scenarios are summarized based on the same initial state of an exemplary keyhole and melt pool and potentially culminate in the formation of:

- Bulging pores (see Fig. 15a), that emerge from a bulge at the rear keyhole wall. The bulge is formed due to the interaction between an upwards-directed melt flow and an expanding keyhole. Once the turning point of the melt is exceeded, the flow direction is reversed. Thus, a high dynamic pressure generated by the melt flow enforces the constriction of this bulge resulting in a pore (required time < 2 ms). The detached pore freezes along the solidification line of the narrow melt pool.
- Spiking pores (see Fig. 15b), that are a product of the spiking phenomenon which leads to a significant increase in keyhole penetration depth occurring at short times (< 0.1 ms). The fast drilling process in turn induces a micro vortex pushing against the kink of the newly formed keyhole tip. This effect is causing

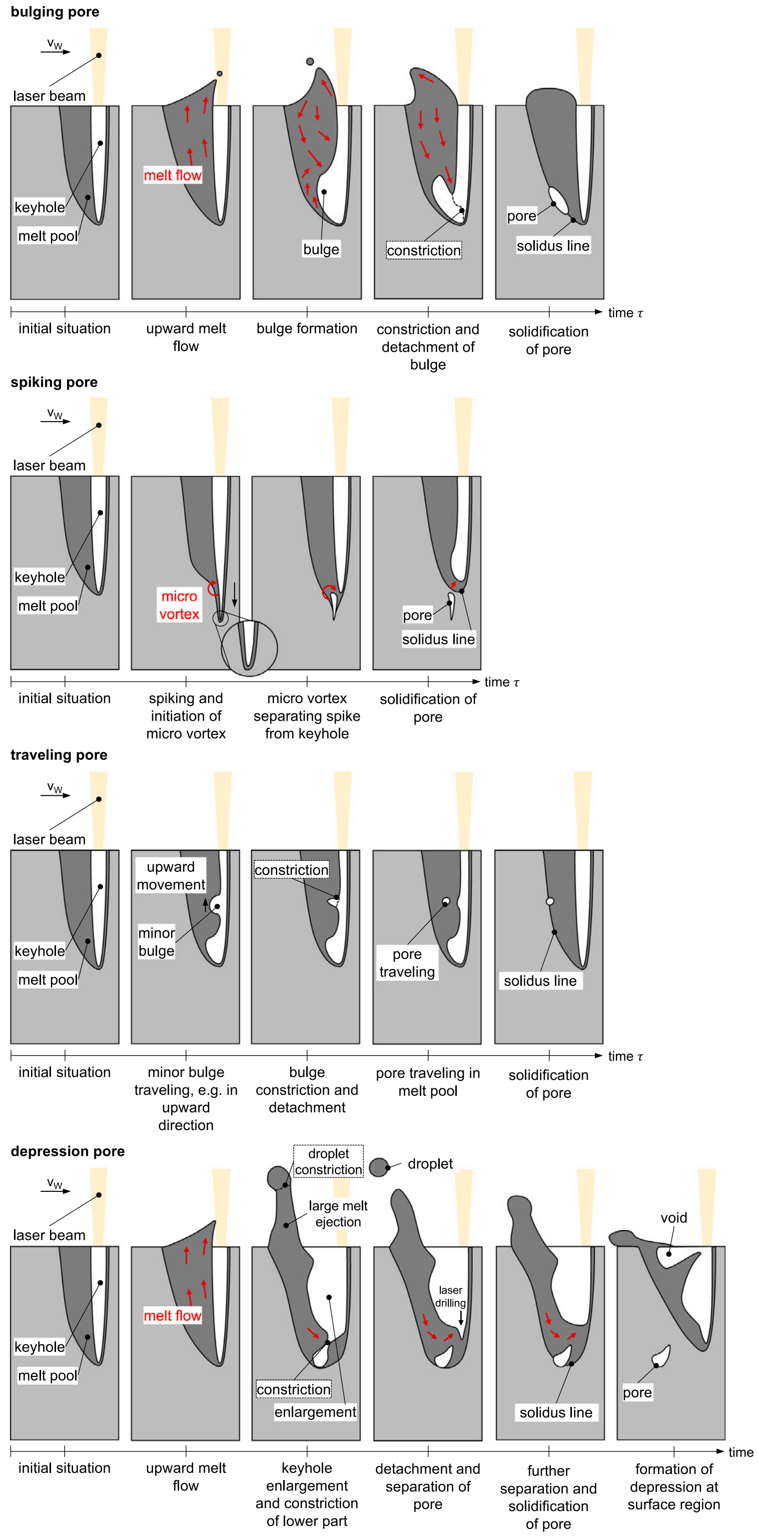

Fig. 15. Schematic illustration of pore formation mechanisms: bulging, spiking, traveling and depression pore.

a detachment of the tip from the keyhole which is followed by immediate solidification of the surrounding material.

- Traveling pores (see Fig. 15c), that originate in local evaporation followed by subsequent heterogeneous vapor flow reinforcing the generation of chaotic and unstable bulges. The fluctuating shape of these bulges are prone to detach minor pores that travel within the melt pool and either get captured through solidification or re-absorbed by the keyhole. This mechanism occurs at the temporal resolution limit of the experiment, i.e., 0.05 ms.
- Depression pores (see Fig. 15d), that are a consequence of substantial upward movement of the melt pool leading to a significantly enlarged aperture of the keyhole towards the surface. The

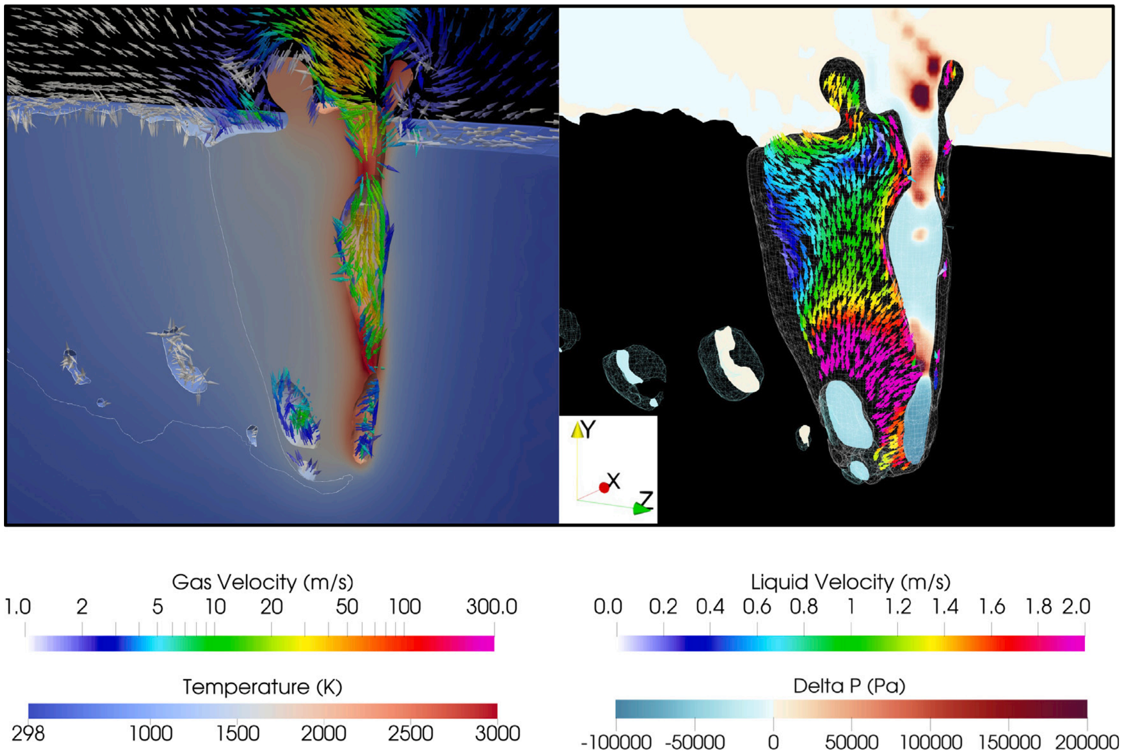

Fig. 16. Exemplary representation of the additional simulation snapshots in the supplementary material. On the left side, temperature is plotted over the condensed phases and velocity over the gaseous phases. On the right side, the pressure difference with respect to ambient pressure is plotted over the gaseous phases and the velocity vectors over the liquid material. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

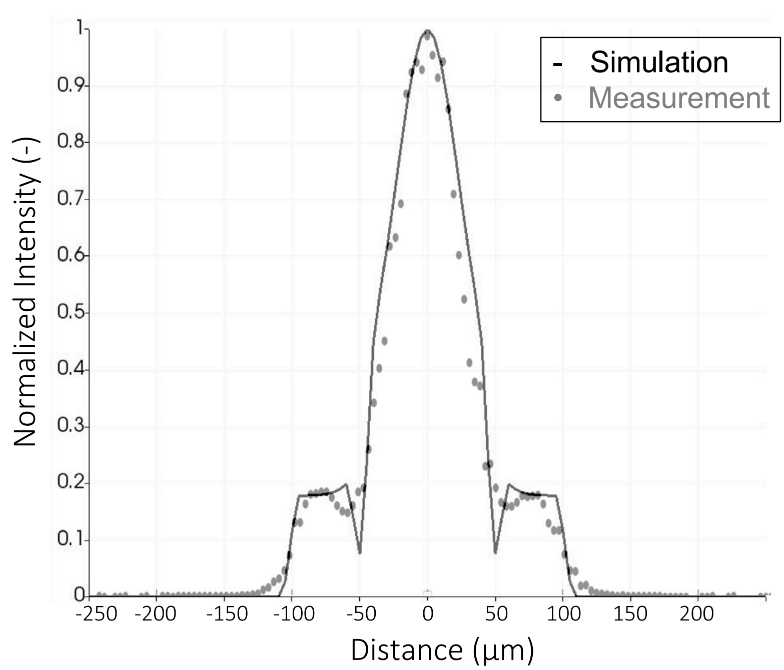

Fig. 17. Normalized simulated (-) and measured (•) intensity profile of the laser beam modeled for the balanced power distribution (50% core and 50% ring intensity, Sim 1b). This plot is recorded at the focus position. The experimentally measured intensity has been recorded at a combined core power of 300 W and ring power of 300 W.

remaining liquid material that is not ejected reverses and exerts high dynamic pressure against the kink of the keyhole bottom detaching a pore. The reformed keyhole is unable to reattach the pore due to the residual melt flow and dissociation through ongoing lasing. Observed times for this formation mechanism were in the range of 1 ms.

The high-speed synchrotron X-ray imaging revealed novel insights into the formation mechanisms of process-related pore formation and also demonstrated a strong effect of concentric core-ring intensity profiles on the reduction of such seam defects in this welding configuration. The exceptional temporal and spatial resolution has facilitated a comprehensive understanding of the dynamic processes involved in laser beam welding of copper. Consequently, our future investigations will focus on examining the temporal evolution of keyhole and melt pool dynamics in response to varying the intensity profiles, particularly in the context of different void formation mechanisms, such as spatter. The in-depth analysis of the keyhole and melt pool over time, their oscillatory behavior, and their correlation with weld defects provides opportunities for the innovation of novel welding processes and the refinement of existing processing strategies.

The hybrid approach of combining high-speed synchrotron X-ray imaging and multi-physics simulation convincingly reveals its potential in extracting comprehensive insights into the underlying process. The symbiotic methodology benefits from its mutual advantages: The simulation profoundly profits from the enhanced possibility for model validation and calibration of elusive material parameters, while the radiographical analysis reaps invaluable perspectives into temperature fields, species distribution, pressure fields, and liquid and vapor flow patterns, provided by the simulation's detailed exposition. Notably, the simulation model used in this study omits common assumptions such as a constant condensation efficiency and a phenomenological recoil pressure model, advancing the state of the art and resulting in results that are significantly closer to reality than any previously reported simulation results in similar combined studies of evaporation-driven keyhole dynamics. The strictly physics-based modeling approach, in combination with in-depth experimental validation, allows for unprecedented confidence in hypotheses gained from the simulation results. Consequently, this work serves as a robust base for further optimizing laser beam welding of copper in a broad range of applications.

CRediT authorship contribution statement

T. Florian: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **K. Schricker:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **C. Zenz:** Writing – review & editing, Writing – original draft, Validation, Software, Investigation, Formal analysis, Conceptualization. **A. Otto:** Writing – review & editing, Supervision, Software, Resources, Methodology, Conceptualization. **L. Schmidt:** Investigation. **C. Diegel:** Investigation. **H. Friedmann:** Investigation. **M. Seibold:** Investigation. **P. Hellwig:** Resources, Investigation. **F. Fröhlich:** Investigation. **F. Nagel:** Methodology, Investigation. **P. Kallage:** Investigation. **M. Buttazzoni:** Writing – review & editing, Software. **A. Rack:** Writing – review & editing, Resources, Investigation. **H. Requardt:** Investigation. **Y. Chen:** Writing – review & editing, Investigation. **J.P. Bergmann:** Writing – review & editing, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 5
Material properties of Cu-ETP: State-independent properties.

Property	T (K)	value	unit	Ref.
Cu-ETP (general)				
molar mass M	–	$63.546 \cdot 10^{-3}$	$kg\,mol^{-1}$	[55]
melting temperature $T_{solidus}$	–	1347	K	[56]
boiling temperature $T_{boiling}$	–	2854	K	
latent heat (fusion) L_{fusion}	–	$205 \cdot 10^{3}$	$m^2\,s^{-2}$	[56]
latent heat (vaporization) L_{vap}	–	$4.75 \cdot 10^{6}$	$m^2\,s^{-2}$	

Table 6
Material properties of Cu-ETP: Solid state.

Property	T (K)	value	unit	Ref.
Cu-ETP (solid)				
density ρ	1347	8320	$kg\,m^{-3}$	[56]
bulk modulus K	300	$100 \cdot 10^{9}$	$kg\,s^{-2}\,m^{-1}$	
specific heat capacity c_p	300	385	$m^2\,s^{-2}\,K^{-1}$	[57]
	973	440	$m^2\,s^{-2}\,K^{-1}$	[57]
thermal conductivity λ	273	390	$kg\,m\,s^{-3}\,K^{-1}$	[56]
	973	300	$kg\,m\,s^{-3}\,K^{-1}$	[56]
	1347	250	$kg\,m\,s^{-3}\,K^{-1}$	
surface energy σ	300	0.1	$kg\,s^{-2}$	
	1339	0.1	$kg\,s^{-2}$	
	1340	1.135	$kg\,s^{-2}$	[58]
refractive index n	–	0.35	–	[59]
extinction coefficient κ	–	7.0	–	[59]
Tait exponent γ	–	7.4	–	

Table 7
Material properties of Cu-ETP: Liquid state.

Property	T (K)	value	unit	Ref.
Cu-ETP (liquid)				
density ρ	1350	7970.8	$kg\,m^{-3}$	[60]
	1700	7724.1	$kg\,m^{-3}$	[60]
bulk modulus K	–	$1 \cdot 10^{9}$	$kg\,s^{-2}\,m^{-1}$	
kinematic viscosity ν	1356	$5.04 \cdot 10^{-7}$	$m^2\,s^{-1}$	[61]
	1500	$4.07 \cdot 10^{-7}$	$m^2\,s^{-1}$	[61]
	1700	$3.24 \cdot 10^{-7}$	$m^2\,s^{-1}$	[61]
	1950	$2.61 \cdot 10^{-7}$	$m^2\,s^{-1}$	[61]
surface energy σ	1347	1.135	$kg\,s^{-2}$	[58]
	2868	0.898	$kg\,s^{-2}$	[58]
specific heat capacity c_p	–	516.86	$m^2\,s^{-1}\,K^{-2}\,K^{-1}$	[57]
thermal conductivity λ	1347	142	$kg\,m\,s^{-3}\,K^{-1}$	[62]
	2854	185	$kg\,m\,s^{-3}\,K^{-1}$	[62]
refractive index n	–	0.96	–	
extinction coefficient κ	–	6.2	–	
Tait exponent γ	–	7.4	–	

Table 8
Material properties of Cu-ETP: Vapor state.

Property	T (K)	value	unit	Ref.
Cu-ETP (vapor)				
density ρ	–	*ideal gas law*		
bulk modulus K	–	$1 \cdot 10^{5}$	$kg\,s^{-2}\,m^{-1}$	
surface energy σ	–	0	$kg\,s^{-2}$	
specific heat capacity c_p	2900	387.9	$m^2\,s^{-2}\,K^{-1}$	[57]
	3500	435.7	$m^2\,s^{-2}\,K^{-1}$	[57]
	4500	498.2	$m^2\,s^{-2}\,K^{-1}$	[57]
thermal conductivity λ	3000	0.15	$kg\,m\,s^{-3}\,K^{-1}$	
	5000	0.24	$kg\,m\,s^{-3}\,K^{-1}$	
kinematic viscosity ν	–	*ideal gas law*		
refractive index n	–	1	–	
extinction coefficient κ	–	0	–	
Tait exponent γ	–	1	–	

Appendix A. Material properties

See Tables 5–9.

Table 9
Material properties of ambient gas.

Property	T (K)	value	unit	Ref.
ambient gas				
molar mass M	–	28.014 $\cdot 10^{-3}$	$\mathrm{kg\,mol^{-1}}$	
density ρ	–	*ideal gas law*		
bulk modulus K	–	1 $\cdot 10^{5}$	$\mathrm{kg\,s^{-2}\,m^{-1}}$	
surface energy σ	–	0	$\mathrm{kg\,s^{-2}}$	
specific heat capacity c_p	300	1041.3	$\mathrm{m^2\,s^{-2}\,K^{-1}}$	[57]
	1800	1270.8	$\mathrm{m^2\,s^{-2}\,K^{-1}}$	[57]
thermal conductivity λ	300	2.597 $\cdot 10^{-2}$	$\mathrm{kg\,m\,s^{-3}\,K^{-1}}$	[57]
	900	6.052 $\cdot 10^{-2}$	$\mathrm{kg\,m\,s^{-3}\,K^{-1}}$	[57]
	1800	10.088 $\cdot 10^{-2}$	$\mathrm{kg\,m\,s^{-3}\,K^{-1}}$	[57]
kinematic viscosity ν	–	*kinetic theory*		
		ν_0=1.593 $\cdot 10^{-5}$	$\mathrm{m^2\,s^{-1}}$	[57]
		ρ_0=1.1233	$\mathrm{kg\,m^{-3}}$	[57]
		T_0=300	K	
refractive index n	–	1	–	
extinction coefficient κ	–	0	–	
Tait exponent γ	–	1	–	

Appendix B. Additional figures

For each described pore formation mechanism, an additional representation of simulation results is added to the supplementary material in order to maximize the information on the mechanisms. Fig. 16 shows an exemplary representation of the snapshot at $\tau = 0.35$ ms within the bulging pore evolution in Fig. 8. On the left hand side, the domain is cut longitudinally along the welding direction. Condensed matter (defined via regions where the sum of liquid and solid volume fraction exceeds 0.5) is colored by temperature. The white line indicates the outline of the solidified weld bead. Arrows indicate the direction (and magnitude via their color) of velocity of both ambient gas and metal vapor within the longitudinal center plane. The velocity ranges from 1 to 300 m/s and is plotted logarithmically. On the right hand side, a longitudinal slice along the welding direction through the center of the laser beam shows the pressure distribution in gaseous regions (defined via regions where the sum of ambient gas and metal vapor exceeds 0.5). Here, not the absolute pressure, but the "Delta P" reduced by the ambient pressure is plotted. In other words, the level around zero indicates pressures around 1 bar. Red and blue colors suggest pressure levels above and below atmospheric pressure, respectively, giving rise to the locations where evaporation and condensation occurs. Furthermore, arrows denote the velocity vectors of the melt flow within the melt pool along the y-z plane from 0 to 2 m/s in a linear scale. These velocity vectors are surrounded by a lightly shaded wireframe of the melt pool outline. Additionally, in order to give a 3-dimensional perspective on the pore morphology, the outline of gaseous phase agglomerations inside the solidified material are also marked by lightly shaded wireframes.

The same representation is used throughout all additional Figures containing simulation results in the supplementary material, with the aim of providing the interested reader with additional quantitative insights.

The supplementary material contains snapshot series as outlined in the discussion 3.2 with the representation style of Fig. 16.

Appendix C. Laser beam profile

Fig. 17 exemplarily shows the normalized intensity profile of the balanced power distribution simulation (50% core and 50% ring intensity, Sim 1b) at the focus position. Therein, as described in Section 2.5, the core intensity is modeled as a superposition of a Gaussian and a top-hat beam profile and the ring is modeled as a top-hat profile. This approach is used for all other simulated intensity distributions. As a comparison, the gray dots in Fig. 17 represent the respective measured values also at balanced power distribution (300 W core and 300 W ring).

Appendix D. Supplementary data

Furthermore, each X-ray series is complemented by a video clip of the radiography recorded with 20 kHz displayed with 15 fps.

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ijmachtools.2024.104224.

Data availability

Data will be made available on request.

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